APPENDIX TO THE REQUIREMENT GATHERING DOCUMENT (RGD v2.0)

GAP-08 — INCREMENTAL UPDATE v1.1

Appendix-H

Entity Relationship Diagram (ERD) & Database Schema Design

School ERP

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Document Version: 1.1

Date: 24 July 2026

Classification: Confidential — Client & Project Team Only

1. Introduction

1.1 Purpose

This document is Appendix-H to the Requirement Gathering Document (RGD v2.0) for the School ERP System, addressing Gap-08: Entity Relationship Diagram (ERD) and Database Schema Design. It defines the complete logical database architecture — entities, relationships, keys, indexes, and integrity rules — for every entity already approved in Appendix-G, as the primary input to the future Physical Database Design phase.

Version 1.1 change note: this is an incremental documentation-quality pass over v1.0. No entity definitions, Entity IDs, or relationships were changed. Additive enhancements only: business relationship names and explicit cardinality notation (Section 5.1); cleaner, labelled module-wise ERD diagrams (Section 8); a per-entity Index Strategy summary (Section 12.3); Data Volume Estimates and Partition Recommendations (Sections 17.7-17.8); a Database Naming Convention section (2.12); and an ERD Validation Metrics summary (Section 20.0). All existing v1.0 content is preserved unchanged.

1.2 Objectives

- Translate Appendix-G's Data Dictionary into a logical ERD with explicit relationship types, cardinality, and referential-integrity rules.
- Identify every junction table required to resolve Many-to-Many relationships.
- Establish normalization compliance (1NF-3NF/BCNF) for every entity.
- Provide index, security, and performance guidance ahead of Physical Database Design.

1.3 Scope

This document covers logical database design only — entity structure, relationships, keys, and integrity rules. It does not include SQL DDL, CREATE TABLE statements, physical storage engine choices, or API design, per the explicit rules governing this document.

1.4 Reference Documents

- RGD v2.0 — Requirement Gathering Document (baseline business requirements)
- Appendix-A — Business Process Diagrams (BPMN)
- Appendix-B — Requirement Traceability Matrix (RTM)
- Appendix-C v1.1 — Business Rule Catalogue
- Appendix-D — Requirement Dependency Matrix (RDM)
- Appendix-E — Functional Requirement Catalogue
- Appendix-F — Non-Functional Requirement Catalogue
- Appendix-G — Data Dictionary / Data Catalogue (direct data source for this document)

1.5 Assumptions

- A relational database management system (RDBMS) is assumed for logical design purposes, consistent with Appendix-G's stated assumption.

- Entity IDs (ENT-<MOD>-###) from Appendix-G are carried forward unchanged as the canonical cross-reference key.
- Surrogate BIGINT auto-increment primary keys are assumed for all entities, per Appendix-G Section 18 (Assumptions).

1.6 Constraints

- No SQL DDL, CREATE TABLE statements, APIs, or backend code are produced in this document, per explicit governing rules.
- Out-of-scope stub entities (Hostel, Visitor, Complaint, Recruitment, DisciplineRecord, RecruitmentRequisition, Event) are carried forward as stubs only, not expanded, consistent with their treatment in Appendix-G.
- Where information is not defined in the approved documents (e.g., Gender, Religion, Blood Group lookups), this document states "Not Defined in Approved Scope" rather than inventing structure.

2. Database Design Principles

2.1 Normalization Strategy

The logical model targets Third Normal Form (3NF) for all transactional and master entities, with deliberate, documented denormalization limited to JSON-typed fields holding variable-structure, rarely-individually-queried data (e.g., PayrollRun.deductions_json, GradingScheme.grade_band_json). See Section 13 for a per-entity normalization review.

2.2 Naming Standards

Identical to the conventions already established in Appendix-G Naming Standards: PascalCase singular entity names, snake_case plural table names, snake_case column names, <table_singular>_id primary/foreign key naming, and uq_/ck_/idx_ prefixed constraint and index names. Not repeated in full here — see Appendix-G.

2.3 Primary Key Strategy

Every entity uses a single-column BIGINT surrogate primary key (e.g., student_id), never a composite business key, except junction tables (Section 6), which use a composite primary key of the two foreign keys they resolve.

2.4 Foreign Key Strategy

Every foreign key is enforced at the database layer via a FK constraint referencing the parent's surrogate primary key, with the default action RESTRICT (Section 11) unless explicitly noted otherwise in the Relationship Catalogue (Section 5). Polymorphic associations (e.g., User.owner_ref_id) cannot be enforced by a standard single-target FK constraint — see Section 11 for the mitigation approach.

2.5 Surrogate Keys vs. Natural Keys

Surrogate keys are the Primary Key for every entity. Natural/business keys (e.g., Student.admission_number, Invoice.invoice_no) are retained as Candidate Keys with a Unique Constraint, giving human-readable lookup without exposing internal surrogate IDs as the join mechanism.

2.6 Audit Columns

Every entity carries: created_by, created_at, updated_by, updated_at (all referencing User), consistent with Appendix-E Field 24 and Appendix-F NFR-AUDIT-001. These are in addition to entity-specific timestamp fields already listed in each entity's Attribute Catalogue in Appendix-G (e.g., Invoice.due_date is a business field, not an audit field).

2.7 Soft Delete Strategy

No entity supports a hard DELETE under normal operation, consistent with the CRUD Operations field already defined for every entity in Appendix-G (Field 16). Every entity carries is_deleted, deleted_by, and deleted_at fields; a "deleted" record is excluded from standard queries but retained for audit and potential Restore, per Appendix-G's Purge Policy (Field 21: no hard purge without a logged, IT Admin-approved exception).

2.8 Versioning

Every entity carries a version integer column, incremented on every Update, to support optimistic concurrency control — directly mitigating the concurrency-conflict error-handling pattern already defined in Appendix-E Field 37.

2.9 Multi-Session Handling

Every session-scoped entity (Application, SeatAllocation, Exam, Invoice, PromotionRecord, etc.) carries an explicit academic_session_id foreign key rather than relying on date-range inference, so historical sessions remain unambiguously queryable even as AcademicSession.status transitions from Active to Closed to Archived.

2.10 Academic Year Strategy

AcademicSession (ENT-ACAD-001) is the single temporal anchor for the entire schema. No entity infers "current year" from a system date; every session-scoped entity's academic_session_id is explicit and immutable once set, preserving historical accuracy even if the current date's default session configuration later changes.

2.11 Institute Isolation Strategy

Not Defined in Approved Scope — RGD v2.0 and Appendix-G do not define multi-institute/multi-tenant requirements; the current scope (RGD v2.0 Section 26 Pending Decisions) has not confirmed whether the platform must support genuinely separate institutes versus multiple campuses of one institute. This logical model assumes single-institute, and multi-campus support (if confirmed) would extend via an institute_id/campus_id column pattern — flagged here as a Future Expansion item (Section 18), not designed in detail without client confirmation.

2.12 Database Naming Convention (v1.1)

Object	Convention	Example
Table Names	snake_case, plural — identical to Appendix-G Naming Standards	students, invoices, marks_records
Primary Keys	<table_singular>_id, surrogate BIGINT	student_id, invoice_id
Foreign Keys	<referenced_table_singular>_id (or _ref_id for polymorphic)	class_id, owner_ref_id
Indexes	idx_<table>_<column(s)>	idx_invoices_student_id
Unique Constraints	uq_<table>_<column(s)>	uq_students_admission_number

3. Entity Relationship Overview

3.1 Notation Legend

Symbol / Notation	Meaning
—▶️	One-to-Many (parent to child), read in the arrow's direction
—(M:N)—▶️	Many-to-Many, resolved via a junction table (Section 6)
1:1	One-to-One relationship
Mandatory	The child record cannot exist without the referenced parent (FK is NOT NULL)
Optional	The child record may exist without the referenced parent (FK is NULLable)
Identifying Relationship	Child's Primary Key includes the parent's key (used only in junction tables, Section 6)
Non-Identifying Relationship	Child has its own independent surrogate Primary Key and merely references the parent via FK (the default for all entities in this schema except junction tables)

3.2 High-Level ERD

Simplified view showing the primary data-flow backbone across modules. Full per-entity relationships are in the Relationship Catalogue (Section 5).

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                              ┌──────────────┐
                              │ AcademicSession │  (temporal anchor for nearly all data)
                              └──────┬───────┘
                                     │
          ┌─────────────┬────────────┼────────────┬────────────┐
          ▼             ▼            ▼            ▼            ▼
     ┌─────────┐   ┌──────────┐  ┌──────────┐  ┌──────────┐  ┌──────────┐
     │Application│─▶│ Student  │─▶│AttendanceRec│ │MarksRecord│ │ Invoice  │
     └─────────┘   └────┬─────┘  └──────────┘  └──────────┘  └────┬─────┘
                        │                                          │
          ┌─────────────┼────────────┬────────────┐                ▼
          ▼             ▼            ▼            ▼          ┌──────────┐
     ┌──────────┐  ┌──────────┐  ┌────────────┐ ┌──────────┐  │ Payment  │
     │ BookIssue│  │TransportAl│ │ ScholarshipW │ │ReportCard│  └──────────┘
     └──────────┘  │ location │  │    aiver     │ └──────────┘
                   └──────────┘  └────────────┘

     ┌──────────┐          ┌───────────────┐          ┌──────────┐
     │ Employee │─────────▶│StaffAttendance│─────────▶│ PayrollRun│
     └────┬─────┘          │    Record     │          └──────────┘
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```
          │               └──────────────┘
          ▼
     ┌──────────────┐
     │ TimetableEntry│  (links Section + Subject + Employee)
     └──────────────┘

     User ──► Role ──►(M:N)──► Permission       Guardian ──(M:N)──► Student
     User ──► AuditLog (every module writes here)   Circular/NotificationLog ──► Guardian/Employee
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3.3 Complete System ERD

The complete system ERD is the union of all Module-wise ERDs in Section 8, cross-linked per the Cross-Module Relationships in Section 9. It is not reproduced as a single monolithic diagram here, as a 39-entity system ERD in ASCII form would be illegible; the module-wise decomposition in Section 8 plus the Relationship Catalogue (Section 5) together constitute the complete system ERD in a usable form.

4. Entity Specifications

Full 24-field ERD specification for every approved entity, organized by module. Business/functional description, data type detail, and full attribute catalogues remain of record in Appendix-G (Field/Section references are preserved below) and are not repeated verbatim here to avoid divergence between the two documents.

Module: Cross-Cutting / System (7 Entities)

ENT-SYS-001 — User

1. Entity Name	User
2. Purpose	Central identity record enabling authentication and role-based authorization across all modules.
3. Primary Key	user_id
4. Foreign Keys	role_id → Role; owner_ref_id (polymorphic → Employee/Student/Guardian per owner_type)
5. Parent Entity	Role
6. Child Entities	AuditLog (via performed_by)
7. Relationship Description	Many-to-One with Role; One-to-One with owning Employee/Student/Guardian record
8. Cardinality	Many-to-One with Role; One-to-One with owning Employee/Student/Guardian record
9. Mandatory / Optional	role_id: Mandatory (NOT NULL); owner_ref_id (polymorphic: Mandatory (NOT NULL)
10. Unique Constraints	username; email
11. Indexes (Recommended)	Unique index on username, email; index on role_id
12. Composite Keys (if applicable)	None — single-column surrogate PK (user_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	username
14. Business Keys	username
15. Derived Attributes	None identified
16. Lookup Dependencies	owner_type (Employee/Student/Guardian); status (Active/Locked/Deactivated); failed_login_count (0-5)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Retained for audit purposes for 7 years post-deactivation (Client/Product Decision Required)*
24. Related FR / BR / Data Dictionary ID	FR: N/A (cross-cutting system entity) | BR: BR-HR-002 | Appendix-G Entity ID: ENT-SYS-001

ENT-SYS-002 — Role

1. Entity Name	Role

2. Purpose	Implements role-based access control (RBAC) consistent with Appendix-E Field 38 permission matrices.
3. Primary Key	role_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	User
7. Relationship Description	One-to-Many with User; Many-to-Many with Permission (via role_permission junction)
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	role_name
11. Indexes (Recommended)	Unique index on role_name
12. Composite Keys (if applicable)	None — single-column surrogate PK (role_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	role_name
14. Business Keys	role_name
15. Derived Attributes	None identified
16. Lookup Dependencies	is_system_role (TRUE/FALSE)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent — roles are structural, not transactional
24. Related FR / BR / Data Dictionary ID	FR: N/A (cross-cutting system entity) | BR: BR-RPT-001, BR-FEE-002, BR-COM-002 | Appendix-G Entity ID: ENT-SYS-002

ENT-SYS-003 — Guardian

1. Entity Name	Guardian
2. Purpose	Enables guardian-student linkage (BR-SIS-006) for portal access and notification delivery.
3. Primary Key	guardian_id
4. Foreign Keys	None (owns the M:N link)

5. Parent Entity	None
6. Child Entities	Student (via junction)
7. Relationship Description	Many-to-Many with Student (via student_guardian_link junction table)
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	None (owns the M:N link): Mandatory (NOT NULL)
10. Unique Constraints	None strictly enforced (siblings may share a guardian mobile number; email if provided should be unique)
11. Indexes (Recommended)	Index on mobile_number for lookup
12. Composite Keys (if applicable)	None — single-column surrogate PK (guardian_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	mobile_number (within reason — see Uniqueness note)
14. Business Keys	mobile_number (within reason — see Uniqueness note)
15. Derived Attributes	None identified
16. Lookup Dependencies	relationship (Father/Mother/Guardian/Other); is_primary_contact (TRUE/FALSE)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Retained per Data Retention policy (Client/Product Decision Required) after last linked student exits*
24. Related FR / BR / Data Dictionary ID	FR: FR-06 | BR: BR-SIS-006 | Appendix-G Entity ID: ENT-SYS-003

ENT-SYS-004 — AuditLog

1. Entity Name	AuditLog
2. Purpose	Directly operationalizes Appendix-E Field 24 and Appendix-F NFR-AUDIT-001 across all modules.
3. Primary Key	audit_log_id
4. Foreign Keys	performed_by → User
5. Parent Entity	None
6. Child Entities	None (terminal log)
7. Relationship Description	Many-to-One with User (performed_by); polymorphic reference to any entity via entity_name + record_id

8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	performed_by: Mandatory (NOT NULL)
10. Unique Constraints	None
11. Indexes (Recommended)	Composite index on (entity_name, record_id); index on performed_by; index on performed_at
12. Composite Keys (if applicable)	None — single-column surrogate PK (audit_log_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	None
14. Business Keys	Not Defined in Approved Scope
15. Derived Attributes	None identified
16. Lookup Dependencies	action (Create/Update/Delete/Approve/Reject/Override)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Minimum 7 years for financial/statutory actions (Client/Product Decision Required to confirm exact period)*
24. Related FR / BR / Data Dictionary ID	FR: N/A (cross-cutting system entity) | BR: All immutability-related rules (BR-SIS-004, BR-ATT-002, BR-EXM-003, BR-FEE-001, BR-HR-002, BR-HR-007) | Appendix-G Entity ID: ENT-SYS-004

ENT-SYS-005 — Configuration

1. Entity Name	Configuration
2. Purpose	Operationalizes NFR-CFG-001 (Appx-F) and the 24 configurable items catalogued in Appendix-C v1.1 Section 3.5, avoiding hard-coded values.
3. Primary Key	setting_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	None
7. Relationship Description	None (standalone reference entity, read by all modules)
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	setting_key

11. Indexes (Recommended)	Unique index on setting_key
12. Composite Keys (if applicable)	None — single-column surrogate PK (setting_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	setting_key
14. Business Keys	setting_key
15. Derived Attributes	None identified
16. Lookup Dependencies	data_type (String/Number/Boolean/Date); is_editable (TRUE/FALSE)
17. Master Data Flag	No
18. Transactional Data Flag	No
19. Reference Data Flag	Yes
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent, with full change history via AuditLog
24. Related FR / BR / Data Dictionary ID	*FR: N/A (cross-cutting system entity) | BR: All rules referencing Client/Product Decision Required in Appendix-C v1.1 | Appendix-G Entity ID: ENT-SYS-005*

ENT-SYS-006 — Document

1. Entity Name	Document
2. Purpose	Single storage/reference pattern for all file-based artifacts across modules.
3. Primary Key	document_id
4. Foreign Keys	owner_ref_id (polymorphic → Application/Student/Invoice per owner_type)
5. Parent Entity	Varies by owner_type
6. Child Entities	None
7. Relationship Description	Many-to-One (polymorphic) with owning entity
8. Cardinality	Many-to-One (polymorphic) with owning entity
9. Mandatory / Optional	owner_ref_id (polymorphic: Mandatory (NOT NULL)
10. Unique Constraints	None
11. Indexes (Recommended)	Index on (owner_type, owner_ref_id)
12. Composite Keys (if applicable)	None — single-column surrogate PK (document_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	None

14. Business Keys	Not Defined in Approved Scope
15. Derived Attributes	None identified
16. Lookup Dependencies	owner_type (Application/Student/Invoice/ReportCard); document_type (Birth Certificate/ID Card/Report Card/Receipt/etc.)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Aligned to the owning entity's retention period
24. Related FR / BR / Data Dictionary ID	FR: FR-01, FR-09, FR-20, FR-23 | BR: None Identified | Appendix-G Entity ID: ENT-SYS-006

ENT-SYS-007 — ApprovalRequest

1. Entity Name	ApprovalRequest
2. Purpose	Provides a single, consistent workflow pattern for the many approval/override paths defined across Appendix-C.
3. Primary Key	approval_request_id
4. Foreign Keys	requested_by → User; approver_id → User
5. Parent Entity	Varies by entity_name
6. Child Entities	None
7. Relationship Description	Many-to-One (polymorphic) with the entity/record being approved; Many-to-One with User (requester and approver)
8. Cardinality	Many-to-One (polymorphic) with the entity/record being approved; Many-to-One with User (requester and approver)
9. Mandatory / Optional	requested_by: Mandatory (NOT NULL); approver_id: Mandatory (NOT NULL)
10. Unique Constraints	None
11. Indexes (Recommended)	Index on (entity_name, record_id); index on status
12. Composite Keys (if applicable)	None — single-column surrogate PK (approval_request_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	None
14. Business Keys	Not Defined in Approved Scope
15. Derived Attributes	None identified
16. Lookup Dependencies	status (Pending/Approved/Rejected)

17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Retained per AuditLog-equivalent retention policy (Client/Product Decision Required)*
24. Related FR / BR / Data Dictionary ID	FR: FR-18, FR-25, FR-35, FR-08 | BR: BR-ATT-003, BR-EXM-003, BR-HR-004, BR-SIS-004, BR-FEE-002 | Appendix-G Entity ID: ENT-SYS-007

Module: Admission & Enrollment (2 Entities)

ENT-ADM-001 — Application

1. Entity Name	Application
2. Purpose	Captures and tracks an admission application from inquiry through confirmed enrollment.
3. Primary Key	application_id
4. Foreign Keys	class_applied_id → Class
5. Parent Entity	Class
6. Child Entities	Document (uploaded application documents)
7. Relationship Description	Many-to-One with Class; One-to-One with Student (upon confirmation, FR-02)
8. Cardinality	Many-to-One with Class; One-to-One with Student (upon confirmation, FR-02)
9. Mandatory / Optional	class_applied_id: Mandatory (NOT NULL)
10. Unique Constraints	application_reference_no
11. Indexes (Recommended)	Unique index on application_reference_no; index on status, class_applied_id
12. Composite Keys (if applicable)	None — single-column surrogate PK (application_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	application_reference_no
14. Business Keys	application_reference_no
15. Derived Attributes	None identified
16. Lookup Dependencies	category (General/RTE); status (Submitted/Verified/Shortlisted/Waitlisted/Admitted/Rejected)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Rejected/withdrawn applications retained 2 years (Client/Product Decision Required); Admitted applications retained per Student retention policy*
24. Related FR / BR / Data Dictionary ID	FR: FR-01, FR-02, FR-03, FR-05, FR-05a, FR-05b | BR: BR-ADM-005, BR-ADM-006 | Appendix-G Entity ID: ENT-ADM-001

ENT-ADM-002 — SeatAllocation

1. Entity Name	SeatAllocation

2. Purpose	Enforces BR-ADM-001 (capacity ceiling) and BR-ADM-003 (RTE ceiling) at the data layer.
3. Primary Key	seat_allocation_id
4. Foreign Keys	class_id → Class; academic_session_id → AcademicSession
5. Parent Entity	Class, AcademicSession
6. Child Entities	None
7. Relationship Description	One-to-One with (Class, AcademicSession) pair; One-to-Many with Application via class_applied_id
8. Cardinality	One-to-One with (Class, AcademicSession) pair; One-to-Many with Application via class_applied_id
9. Mandatory / Optional	class_id: Mandatory (NOT NULL); academic_session_id: Mandatory (NOT NULL)
10. Unique Constraints	(class_id, academic_session_id)
11. Indexes (Recommended)	Unique composite index on (class_id, academic_session_id)
12. Composite Keys (if applicable)	None — single-column surrogate PK (seat_allocation_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(class_id, academic_session_id)
14. Business Keys	(class_id, academic_session_id)
15. Derived Attributes	seats_filled and rte_seats_filled are derived counters, incremented by Application confirmation events
16. Lookup Dependencies	None — no enumerated/lookup-constrained attributes
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (historical capacity record)
24. Related FR / BR / Data Dictionary ID	FR: FR-04 | BR: BR-ADM-001, BR-ADM-003, BR-ADM-007, BR-ADM-008 | Appendix-G Entity ID: ENT-ADM-002

Module: Academic Master Data (5 Entities)

ENT-ACAD-001 — AcademicSession

1. Entity Name	AcademicSession
2. Purpose	Anchors every transactional entity to the correct academic year for reporting and historical retention.
3. Primary Key	academic_session_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	Application, SeatAllocation, AttendanceRecord, Exam, Invoice, PromotionRecord, etc.
7. Relationship Description	One-to-Many with nearly every transactional entity
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	session_name; (start_date, end_date) non-overlapping
11. Indexes (Recommended)	Unique index on session_name
12. Composite Keys (if applicable)	None — single-column surrogate PK (academic_session_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	session_name
14. Business Keys	session_name
15. Derived Attributes	None identified
16. Lookup Dependencies	status (Planned/Active/Closed/Archived)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (historical sessions never purged, only archived)
24. Related FR / BR / Data Dictionary ID	FR: N/A (implicit foundational master data across all FRs) | BR: BR-SIS-001 | Appendix-G Entity ID: ENT-ACAD-001

ENT-ACAD-002 — Class

1. Entity Name	Class

2. Purpose	Structural master data referenced by Admission, SIS, Timetable, Fee, and Examination.
3. Primary Key	class_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	Section
7. Relationship Description	One-to-Many with Section; One-to-Many with Application, Student (via Section), FeeStructure
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	class_name
11. Indexes (Recommended)	Unique index on class_name
12. Composite Keys (if applicable)	None — single-column surrogate PK (class_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	class_name
14. Business Keys	class_name
15. Derived Attributes	None identified
16. Lookup Dependencies	None — no enumerated/lookup-constrained attributes
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent
24. Related FR / BR / Data Dictionary ID	FR: FR-06, FR-14, FR-22 | BR: None Identified | Appendix-G Entity ID: ENT-ACAD-002

ENT-ACAD-003 — Section

1. Entity Name	Section
2. Purpose	Enables class-size management and homeroom-level operations.
3. Primary Key	section_id
4. Foreign Keys	class_id → Class

5. Parent Entity	Class
6. Child Entities	Student, TimetableEntry
7. Relationship Description	Many-to-One with Class; One-to-Many with Student, TimetableEntry
8. Cardinality	Many-to-One with Class; One-to-Many with Student, TimetableEntry
9. Mandatory / Optional	class_id: Mandatory (NOT NULL)
10. Unique Constraints	(class_id, section_name)
11. Indexes (Recommended)	Unique composite index on (class_id, section_name)
12. Composite Keys (if applicable)	None — single-column surrogate PK (section_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(class_id, section_name)
14. Business Keys	(class_id, section_name)
15. Derived Attributes	None identified
16. Lookup Dependencies	None — no enumerated/lookup-constrained attributes
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent
24. Related FR / BR / Data Dictionary ID	FR: FR-06, FR-14 | BR: BR-SIS-005 | Appendix-G Entity ID: ENT-ACAD-003

ENT-ACAD-004 — Subject

1. Entity Name	Subject
2. Purpose	Structural master data underpinning timetable scheduling and marks entry.
3. Primary Key	subject_id
4. Foreign Keys	None (class mapping via ClassSubjectMap junction, described in Relationships)
5. Parent Entity	None
6. Child Entities	TimetableEntry, MarksRecord
7. Relationship Description	Many-to-Many with Class (via class_subject_map junction); One-to-Many with TimetableEntry, MarksRecord

8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	None (class mapping via ClassSubjectMap junction, described in Relationships): Mandatory (NOT NULL)
10. Unique Constraints	subject_code
11. Indexes (Recommended)	Unique index on subject_code
12. Composite Keys (if applicable)	None — single-column surrogate PK (subject_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	subject_code
14. Business Keys	subject_code
15. Derived Attributes	None identified
16. Lookup Dependencies	None — no enumerated/lookup-constrained attributes
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent
24. Related FR / BR / Data Dictionary ID	FR: FR-17, FR-14 | BR: None Identified | Appendix-G Entity ID: ENT-ACAD-004

ENT-ACAD-005 — GradingScheme

1. Entity Name	GradingScheme
2. Purpose	Ensures results are computed per the school's confirmed board affiliation (BR-EXM-007).
3. Primary Key	grading_scheme_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	Exam
7. Relationship Description	One-to-Many with Exam
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	scheme_name

11. Indexes (Recommended)	Unique index on scheme_name
12. Composite Keys (if applicable)	None — single-column surrogate PK (grading_scheme_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	scheme_name
14. Business Keys	scheme_name
15. Derived Attributes	None identified
16. Lookup Dependencies	board_type (CBSE/ICSE/State Board)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (historical schemes must remain available for closed-year report cards)
24. Related FR / BR / Data Dictionary ID	FR: FR-17, FR-19 | BR: BR-EXM-007 | Appendix-G Entity ID: ENT-ACAD-005

Module: Student Information Management (1 Entity)

ENT-SIS-001 — Student

1. Entity Name	Student
2. Purpose	Single system of record consumed by every other module (Appendix-D Critical Dependency List).
3. Primary Key	student_id
4. Foreign Keys	section_id → Section; application_id → Application
5. Parent Entity	Section
6. Child Entities	AttendanceRecord, MarksRecord, Invoice, BookIssue, TransportAllocation, PromotionRecord, ReportCard, Document
7. Relationship Description	One-to-One with Application; Many-to-One with Section; Many-to-Many with Guardian; One-to-Many with AttendanceRecord, MarksRecord, Invoice, BookIssue, TransportAllocation, PromotionRecord
8. Cardinality	One-to-One with Application; Many-to-One with Section; Many-to-Many with Guardian; One-to-Many with AttendanceRecord, MarksRecord, Invoice, BookIssue, TransportAllocation, PromotionRecord
9. Mandatory / Optional	section_id: Mandatory (NOT NULL); application_id: Mandatory (NOT NULL)
10. Unique Constraints	admission_number; aadhaar_number (where provided)
11. Indexes (Recommended)	Unique index on admission_number; unique index on aadhaar_number; index on section_id; search index on full_name (NFR-SRCH-001)
12. Composite Keys (if applicable)	None — single-column surrogate PK (student_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	admission_number; aadhaar_number
14. Business Keys	admission_number; aadhaar_number
15. Derived Attributes	None identified
16. Lookup Dependencies	category (General/RTE); status (Draft/Active/Promoted/Exited/Archived)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Retained per Data Retention policy (Client/Product Decision Required), typically several years post-exit for certificate re-issuance*
24. Related FR / BR / Data Dictionary ID	FR: FR-06, FR-07, FR-08, FR-09 | BR: BR-SIS-002, BR-SIS-003, BR-SIS-005, BR-SIS-006 | Appendix-G Entity ID: ENT-SIS-001

Module: Attendance Management (2 Entities)

ENT-ATT-001 — AttendanceRecord

1. Entity Name	AttendanceRecord
2. Purpose	Feeds notifications, examination eligibility, and reporting.
3. Primary Key	attendance_record_id
4. Foreign Keys	student_id → Student; timetable_entry_id → TimetableEntry
5. Parent Entity	Student, TimetableEntry
6. Child Entities	None
7. Relationship Description	Many-to-One with Student; Many-to-One with TimetableEntry
8. Cardinality	Many-to-One with Student; Many-to-One with TimetableEntry
9. Mandatory / Optional	student_id: Mandatory (NOT NULL); timetable_entry_id: Mandatory (NOT NULL)
10. Unique Constraints	(student_id, timetable_entry_id, attendance_date)
11. Indexes (Recommended)	Unique composite index on (student_id, timetable_entry_id, attendance_date)
12. Composite Keys (if applicable)	None — single-column surrogate PK (attendance_record_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(student_id, timetable_entry_id, attendance_date)
14. Business Keys	(student_id, timetable_entry_id, attendance_date)
15. Derived Attributes	None directly, but is_locked derives attendance-percentage rollups consumed by FR-13
16. Lookup Dependencies	state (Present/Absent/Late); is_locked (TRUE/FALSE)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	10 million+ record scale per NFR-SCAL-005; archived per NFR-ARC-001 after academic session close
24. Related FR / BR / Data Dictionary ID	FR: FR-10, FR-11, FR-13 | BR: BR-ATT-001, BR-ATT-002, BR-ATT-003, BR-ATT-007 | Appendix-G Entity ID: ENT-ATT-001

ENT-ATT-002 — StaffAttendanceRecord

1. Entity Name	StaffAttendanceRecord

2. Purpose	Feeds payroll calculation (BR-HR-001).
3. Primary Key	staff_attendance_id
4. Foreign Keys	employee_id → Employee
5. Parent Entity	Employee
6. Child Entities	None
7. Relationship Description	Many-to-One with Employee
8. Cardinality	Many-to-One with Employee
9. Mandatory / Optional	employee_id: Mandatory (NOT NULL)
10. Unique Constraints	(employee_id, attendance_date)
11. Indexes (Recommended)	Unique composite index on (employee_id, attendance_date)
12. Composite Keys (if applicable)	None — single-column surrogate PK (staff_attendance_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(employee_id, attendance_date)
14. Business Keys	(employee_id, attendance_date)
15. Derived Attributes	None identified
16. Lookup Dependencies	state (Present/On Leave/Unauthorized); is_reconciled (TRUE/FALSE)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Aligned with PayrollRun retention (statutory, Client/Product Decision Required)*
24. Related FR / BR / Data Dictionary ID	FR: FR-12 | BR: BR-ATT-005, BR-HR-001 | Appendix-G Entity ID: ENT-ATT-002

Module: Timetable & Scheduling (1 Entity)

ENT-TT-001 — TimetableEntry

1. Entity Name	TimetableEntry
2. Purpose	Structural backbone for Attendance and Examination scheduling.
3. Primary Key	timetable_entry_id
4. Foreign Keys	section_id → Section; subject_id → Subject; employee_id → Employee (teacher)
5. Parent Entity	Section, Subject, Employee
6. Child Entities	AttendanceRecord, MarksRecord (via Exam/Subject)
7. Relationship Description	Many-to-One with Section, Subject, Employee
8. Cardinality	Many-to-One with Section, Subject, Employee
9. Mandatory / Optional	section_id: Mandatory (NOT NULL); subject_id: Mandatory (NOT NULL); employee_id: Mandatory (NOT NULL)
10. Unique Constraints	(section_id, day_of_week, period_no); (employee_id, day_of_week, period_no) — enforces BR-TT-001
11. Indexes (Recommended)	Unique composite index on (section_id, day_of_week, period_no); unique composite index on (employee_id, day_of_week, period_no)
12. Composite Keys (if applicable)	None — single-column surrogate PK (timetable_entry_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(section_id, day_of_week, period_no)
14. Business Keys	(section_id, day_of_week, period_no)
15. Derived Attributes	None identified
16. Lookup Dependencies	*day_of_week (Mon-Sat); period_no (1-8 (Client/Product Decision Required for max periods/day)); status (Draft/Published)*
17. Master Data Flag	No
18. Transactional Data Flag	No
19. Reference Data Flag	Yes
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Historical versions retained for the academic session; archived thereafter
24. Related FR / BR / Data Dictionary ID	FR: FR-14, FR-15, FR-16 | BR: BR-TT-001, BR-TT-002, BR-TT-003, BR-TT-005, BR-TT-006 | Appendix-G Entity ID: ENT-TT-001

Module: Examination & Result Processing (4 Entities)

ENT-EXM-001 — Exam

1. Entity Name	Exam
2. Purpose	Structural basis for marks entry and result calculation.
3. Primary Key	exam_id
4. Foreign Keys	academic_session_id → AcademicSession; grading_scheme_id → GradingScheme; class_id → Class
5. Parent Entity	AcademicSession, Class
6. Child Entities	MarksRecord, ReportCard
7. Relationship Description	Many-to-One with AcademicSession, GradingScheme, Class; One-to-Many with MarksRecord
8. Cardinality	Many-to-One with AcademicSession, GradingScheme, Class; One-to-Many with MarksRecord
9. Mandatory / Optional	academic_session_id: Mandatory (NOT NULL); grading_scheme_id: Mandatory (NOT NULL); class_id: Mandatory (NOT NULL)
10. Unique Constraints	(class_id, exam_name, academic_session_id)
11. Indexes (Recommended)	Unique composite index on (class_id, exam_name, academic_session_id)
12. Composite Keys (if applicable)	None — single-column surrogate PK (exam_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(class_id, exam_name, academic_session_id)
14. Business Keys	(class_id, exam_name, academic_session_id)
15. Derived Attributes	None identified
16. Lookup Dependencies	status (Configured/Active/Locked/Closed)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (historical exam record)
24. Related FR / BR / Data Dictionary ID	FR: FR-17 | BR: BR-EXM-005, BR-EXM-007 | Appendix-G Entity ID: ENT-EXM-001

ENT-EXM-002 — MarksRecord

1. Entity Name	MarksRecord

2. Purpose	Source data for grade/rank/GPA calculation.
3. Primary Key	marks_record_id
4. Foreign Keys	exam_id → Exam; student_id → Student; subject_id → Subject
5. Parent Entity	Exam, Student, Subject
6. Child Entities	None
7. Relationship Description	Many-to-One with Exam, Student, Subject
8. Cardinality	Many-to-One with Exam, Student, Subject
9. Mandatory / Optional	exam_id: Mandatory (NOT NULL); student_id: Mandatory (NOT NULL); subject_id: Mandatory (NOT NULL)
10. Unique Constraints	(exam_id, student_id, subject_id)
11. Indexes (Recommended)	Unique composite index on (exam_id, student_id, subject_id)
12. Composite Keys (if applicable)	None — single-column surrogate PK (marks_record_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(exam_id, student_id, subject_id)
14. Business Keys	(exam_id, student_id, subject_id)
15. Derived Attributes	None identified
16. Lookup Dependencies	is_flagged (TRUE/FALSE); is_locked (TRUE/FALSE)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	5 million+ record scale per NFR-SCAL-006; archived per NFR-ARC-001
24. Related FR / BR / Data Dictionary ID	FR: FR-18, FR-19 | BR: BR-EXM-002, BR-EXM-003, BR-EXM-006 | Appendix-G Entity ID: ENT-EXM-002

ENT-EXM-003 — ReportCard

1. Entity Name	ReportCard
2. Purpose	Compiles calculated results into the parent/student-facing document.
3. Primary Key	report_card_id
4. Foreign Keys	student_id → Student; exam_id → Exam

5. Parent Entity	Student, Exam
6. Child Entities	Document
7. Relationship Description	Many-to-One with Student, Exam; One-to-One with Document (generated PDF)
8. Cardinality	Many-to-One with Student, Exam; One-to-One with Document (generated PDF)
9. Mandatory / Optional	student_id: Mandatory (NOT NULL); exam_id: Mandatory (NOT NULL)
10. Unique Constraints	(student_id, exam_id)
11. Indexes (Recommended)	Unique composite index on (student_id, exam_id)
12. Composite Keys (if applicable)	None — single-column surrogate PK (report_card_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(student_id, exam_id)
14. Business Keys	(student_id, exam_id)
15. Derived Attributes	gpa and class_rank are derived from locked MarksRecord values via the grade-calculation engine (FR-19), not directly entered
16. Lookup Dependencies	is_published (TRUE/FALSE)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (academic record)
24. Related FR / BR / Data Dictionary ID	FR: FR-20, FR-21 | BR: BR-EXM-001 | Appendix-G Entity ID: ENT-EXM-003

ENT-EXM-004 — PromotionRecord

1. Entity Name	PromotionRecord
2. Purpose	Enforces and documents BR-SIS-001 (academic/fee closure precondition).
3. Primary Key	promotion_record_id
4. Foreign Keys	student_id → Student; from_session_id → AcademicSession; to_session_id → AcademicSession; from_class_id → Class; to_class_id → Class
5. Parent Entity	Student
6. Child Entities	None
7. Relationship Description	Many-to-One with Student; Many-to-One with AcademicSession (twice, from/to); Many-to-One with Class (twice, from/to)

8. Cardinality	Many-to-One with Student; Many-to-One with AcademicSession (twice, from/to); Many-to-One with Class (twice, from/to)
9. Mandatory / Optional	student_id: Mandatory (NOT NULL); from_session_id: Mandatory (NOT NULL); to_session_id: Mandatory (NOT NULL); from_class_id: Mandatory (NOT NULL); to_class_id: Mandatory (NOT NULL)
10. Unique Constraints	(student_id, from_session_id)
11. Indexes (Recommended)	Unique composite index on (student_id, from_session_id)
12. Composite Keys (if applicable)	None — single-column surrogate PK (promotion_record_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(student_id, from_session_id)
14. Business Keys	(student_id, from_session_id)
15. Derived Attributes	None identified
16. Lookup Dependencies	academic_closure_confirmed (TRUE/FALSE); fee_closure_confirmed (TRUE/FALSE)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (historical academic record, part of FR-08)
24. Related FR / BR / Data Dictionary ID	FR: FR-08 | BR: BR-SIS-001 | Appendix-G Entity ID: ENT-EXM-004

Module: Fee Collection (5 Entities)

ENT-FEE-001 — FeeHead

1. Entity Name	FeeHead
2. Purpose	Structural master data for fee configuration.
3. Primary Key	fee_head_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	FeeStructure
7. Relationship Description	One-to-Many with FeeStructure
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	fee_head_name
11. Indexes (Recommended)	Unique index on fee_head_name
12. Composite Keys (if applicable)	None — single-column surrogate PK (fee_head_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	fee_head_name
14. Business Keys	fee_head_name
15. Derived Attributes	None identified
16. Lookup Dependencies	is_taxable (TRUE/FALSE)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent
24. Related FR / BR / Data Dictionary ID	FR: FR-22 | BR: None Identified | Appendix-G Entity ID: ENT-FEE-001

ENT-FEE-002 — FeeStructure

1. Entity Name	FeeStructure

2. Purpose	Billing foundation for automated invoice generation.
3. Primary Key	fee_structure_id
4. Foreign Keys	class_id → Class; fee_head_id → FeeHead; academic_session_id → AcademicSession
5. Parent Entity	Class, FeeHead, AcademicSession
6. Child Entities	Invoice (line items)
7. Relationship Description	Many-to-One with Class, FeeHead, AcademicSession
8. Cardinality	Many-to-One with Class, FeeHead, AcademicSession
9. Mandatory / Optional	class_id: Mandatory (NOT NULL); fee_head_id: Mandatory (NOT NULL); academic_session_id: Mandatory (NOT NULL)
10. Unique Constraints	(class_id, fee_head_id, academic_session_id, category)
11. Indexes (Recommended)	Unique composite index on (class_id, fee_head_id, academic_session_id, category)
12. Composite Keys (if applicable)	None — single-column surrogate PK (fee_structure_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(class_id, fee_head_id, academic_session_id, category)
14. Business Keys	(class_id, fee_head_id, academic_session_id, category)
15. Derived Attributes	None identified
16. Lookup Dependencies	category (General/RTE)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (needed to explain historical invoices)
24. Related FR / BR / Data Dictionary ID	FR: FR-22 | BR: None Identified | Appendix-G Entity ID: ENT-FEE-002

ENT-FEE-003 — Invoice

1. Entity Name	Invoice
2. Purpose	Core financial document driving payment collection.
3. Primary Key	invoice_id
4. Foreign Keys	student_id → Student; academic_session_id → AcademicSession

5. Parent Entity	Student, AcademicSession
6. Child Entities	Payment
7. Relationship Description	Many-to-One with Student, AcademicSession; One-to-Many with Payment
8. Cardinality	Many-to-One with Student, AcademicSession; One-to-Many with Payment
9. Mandatory / Optional	student_id: Mandatory (NOT NULL); academic_session_id: Mandatory (NOT NULL)
10. Unique Constraints	invoice_no
11. Indexes (Recommended)	Unique index on invoice_no; index on student_id, status, due_date
12. Composite Keys (if applicable)	None — single-column surrogate PK (invoice_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	invoice_no
14. Business Keys	invoice_no
15. Derived Attributes	total_amount is derived as the sum of applicable FeeStructure line items minus any ScholarshipWaiver amounts
16. Lookup Dependencies	status (Unpaid/Paid/Partially Paid/Defaulter/Cancelled); is_locked (TRUE/FALSE)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*500,000+ transaction scale per NFR-SCAL-007; statutory financial retention period (Client/Product Decision Required)*
24. Related FR / BR / Data Dictionary ID	FR: FR-23, FR-25, FR-26 | BR: BR-FEE-001 | Appendix-G Entity ID: ENT-FEE-003

ENT-FEE-004 — Payment

1. Entity Name	Payment
2. Purpose	Records fee collection and gateway transaction linkage.
3. Primary Key	payment_id
4. Foreign Keys	invoice_id → Invoice
5. Parent Entity	Invoice
6. Child Entities	None
7. Relationship Description	Many-to-One with Invoice

8. Cardinality	Many-to-One with Invoice
9. Mandatory / Optional	invoice_id: Mandatory (NOT NULL)
10. Unique Constraints	gateway_transaction_ref (where present, per BR-FEE-006)
11. Indexes (Recommended)	Unique index on gateway_transaction_ref; index on invoice_id
12. Composite Keys (if applicable)	None — single-column surrogate PK (payment_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	gateway_transaction_ref
14. Business Keys	gateway_transaction_ref
15. Derived Attributes	None identified
16. Lookup Dependencies	payment_mode (Online/Cash/Cheque/Bank Transfer); status (Success/Failed/Refunded/Voided)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Statutory financial retention period (Client/Product Decision Required)*
24. Related FR / BR / Data Dictionary ID	FR: FR-24 | BR: BR-FEE-002, BR-FEE-006 | Appendix-G Entity ID: ENT-FEE-004

ENT-FEE-005 — ScholarshipWaiver

1. Entity Name	ScholarshipWaiver
2. Purpose	Tracks waived amounts separately for RTE reimbursement reporting (FR-26b).
3. Primary Key	scholarship_waiver_id
4. Foreign Keys	student_id → Student; fee_head_id → FeeHead
5. Parent Entity	Student, FeeHead
6. Child Entities	None
7. Relationship Description	Many-to-One with Student, FeeHead
8. Cardinality	Many-to-One with Student, FeeHead
9. Mandatory / Optional	student_id: Mandatory (NOT NULL); fee_head_id: Mandatory (NOT NULL)
10. Unique Constraints	None

11. Indexes (Recommended)	Index on student_id
12. Composite Keys (if applicable)	None — single-column surrogate PK (scholarship_waiver_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	None
14. Business Keys	Not Defined in Approved Scope
15. Derived Attributes	None identified
16. Lookup Dependencies	waiver_type (RTE/Merit/Sibling/Staff-Ward)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Aligned with Invoice retention (statutory)
24. Related FR / BR / Data Dictionary ID	FR: FR-26b | BR: BR-FEE-005 | Appendix-G Entity ID: ENT-FEE-005

Module: Library Management (2 Entities)

ENT-LIB-001 — Book

1. Entity Name	Book
2. Purpose	Structural master data for issue/return processing and self-service search.
3. Primary Key	book_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	BookIssue
7. Relationship Description	One-to-Many with BookIssue
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	barcode
11. Indexes (Recommended)	Unique index on barcode; search index on title, author (NFR-SRCH-001)
12. Composite Keys (if applicable)	None — single-column surrogate PK (book_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	barcode
14. Business Keys	barcode
15. Derived Attributes	None identified
16. Lookup Dependencies	classification (Circulating/Reference); is_available (TRUE/FALSE)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (catalog history)
24. Related FR / BR / Data Dictionary ID	FR: FR-27, FR-29 | BR: BR-LIB-004 | Appendix-G Entity ID: ENT-LIB-001

ENT-LIB-002 — BookIssue

1. Entity Name	BookIssue

2. Purpose	Tracks circulation and overdue fine calculation.
3. Primary Key	book_issue_id
4. Foreign Keys	book_id → Book; borrower_ref_id (polymorphic → Student/Employee per borrower_type)
5. Parent Entity	Book
6. Child Entities	None
7. Relationship Description	Many-to-One with Book; Many-to-One (polymorphic) with Student or Employee
8. Cardinality	Many-to-One with Book; Many-to-One (polymorphic) with Student or Employee
9. Mandatory / Optional	book_id: Mandatory (NOT NULL); borrower_ref_id (polymorphic: Mandatory (NOT NULL)
10. Unique Constraints	None
11. Indexes (Recommended)	Index on book_id; index on (borrower_type, borrower_ref_id); index on due_date (for overdue scans)
12. Composite Keys (if applicable)	None — single-column surrogate PK (book_issue_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	None
14. Business Keys	Not Defined in Approved Scope
15. Derived Attributes	fine_amount is derived from (return_date − due_date) × configured per-day rate when overdue
16. Lookup Dependencies	borrower_type (Student/Employee)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (borrowing history)
24. Related FR / BR / Data Dictionary ID	FR: FR-28 | BR: BR-LIB-001, BR-LIB-002, BR-LIB-005 | Appendix-G Entity ID: ENT-LIB-002

Module: Transport Management (3 Entities)

ENT-TRN-001 — Route

1. Entity Name	Route
2. Purpose	Structural master data for transport allocation.
3. Primary Key	route_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	TransportAllocation
7. Relationship Description	One-to-Many with TransportAllocation; One-to-One (or One-to-Many across sessions) with Vehicle
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	route_name
11. Indexes (Recommended)	Unique index on route_name
12. Composite Keys (if applicable)	None — single-column surrogate PK (route_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	route_name
14. Business Keys	route_name
15. Derived Attributes	None identified
16. Lookup Dependencies	None — no enumerated/lookup-constrained attributes
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent
24. Related FR / BR / Data Dictionary ID	FR: FR-30 | BR: BR-TRN-001 | Appendix-G Entity ID: ENT-TRN-001

ENT-TRN-002 — Vehicle

1. Entity Name	Vehicle

2. Purpose	Physical asset backing route capacity and live tracking.
3. Primary Key	vehicle_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	Route
7. Relationship Description	One-to-Many with Route (across time, though typically one active at a time)
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	registration_no
11. Indexes (Recommended)	Unique index on registration_no
12. Composite Keys (if applicable)	None — single-column surrogate PK (vehicle_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	registration_no
14. Business Keys	registration_no
15. Derived Attributes	None identified
16. Lookup Dependencies	None — no enumerated/lookup-constrained attributes
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent (safety audit history)
24. Related FR / BR / Data Dictionary ID	FR: FR-30, FR-31 | BR: BR-TRN-001, BR-TRN-006 | Appendix-G Entity ID: ENT-TRN-002

ENT-TRN-003 — TransportAllocation

1. Entity Name	TransportAllocation
2. Purpose	Links a student to transport service and drives fee linkage.
3. Primary Key	transport_allocation_id
4. Foreign Keys	student_id → Student; route_id → Route

5. Parent Entity	Student, Route
6. Child Entities	None
7. Relationship Description	Many-to-One with Student, Route
8. Cardinality	Many-to-One with Student, Route
9. Mandatory / Optional	student_id: Mandatory (NOT NULL); route_id: Mandatory (NOT NULL)
10. Unique Constraints	student_id where status='Active'
11. Indexes (Recommended)	Unique index on student_id where status is Active
12. Composite Keys (if applicable)	None — single-column surrogate PK (transport_allocation_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	student_id (active allocation must be unique per BR-TRN-002)
14. Business Keys	student_id (active allocation must be unique per BR-TRN-002)
15. Derived Attributes	None identified
16. Lookup Dependencies	status (Active/Waitlisted/De-allocated)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Aligned with Student retention
24. Related FR / BR / Data Dictionary ID	FR: FR-30, FR-32 | BR: BR-TRN-002, BR-TRN-004 | Appendix-G Entity ID: ENT-TRN-003

Module: HR & Payroll (5 Entities)

ENT-HR-001 — Employee

1. Entity Name	Employee
2. Purpose	Single system of record for all staff-related modules.
3. Primary Key	employee_id
4. Foreign Keys	department_id → Department; designation_id → Designation
5. Parent Entity	Department, Designation
6. Child Entities	StaffAttendanceRecord, PayrollRun, LeaveRequest
7. Relationship Description	Many-to-One with Department, Designation; One-to-Many with StaffAttendanceRecord, PayrollRun, LeaveRequest, TimetableEntry (as teacher)
8. Cardinality	Many-to-One with Department, Designation; One-to-Many with StaffAttendanceRecord, PayrollRun, LeaveRequest, TimetableEntry (as teacher)
9. Mandatory / Optional	department_id: Mandatory (NOT NULL); designation_id: Mandatory (NOT NULL)
10. Unique Constraints	employee_code
11. Indexes (Recommended)	Unique index on employee_code; index on department_id, designation_id
12. Composite Keys (if applicable)	None — single-column surrogate PK (employee_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	employee_code
14. Business Keys	employee_code
15. Derived Attributes	None identified
16. Lookup Dependencies	status (Active/Exited)
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Statutory employment record retention period (Client/Product Decision Required)*
24. Related FR / BR / Data Dictionary ID	FR: FR-33 | BR: BR-HR-002, BR-HR-006 | Appendix-G Entity ID: ENT-HR-001

ENT-HR-002 — Department

1. Entity Name	Department

2. Purpose	Structural master data for staff organization.
3. Primary Key	department_id
4. Foreign Keys	None
5. Parent Entity	None
6. Child Entities	Employee
7. Relationship Description	One-to-Many with Employee
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	department_name
11. Indexes (Recommended)	Unique index on department_name
12. Composite Keys (if applicable)	None — single-column surrogate PK (department_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	department_name
14. Business Keys	department_name
15. Derived Attributes	None identified
16. Lookup Dependencies	None — no enumerated/lookup-constrained attributes
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent
24. Related FR / BR / Data Dictionary ID	FR: FR-33 | BR: None Identified | Appendix-G Entity ID: ENT-HR-002

ENT-HR-003 — Designation

1. Entity Name	Designation
2. Purpose	Structural master data for staff role classification.
3. Primary Key	designation_id
4. Foreign Keys	None

5. Parent Entity	None
6. Child Entities	Employee
7. Relationship Description	One-to-Many with Employee
8. Cardinality	N/A — root/reference entity with no parent
9. Mandatory / Optional	N/A — no foreign key dependency
10. Unique Constraints	designation_name
11. Indexes (Recommended)	Unique index on designation_name
12. Composite Keys (if applicable)	None — single-column surrogate PK (designation_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	designation_name
14. Business Keys	designation_name
15. Derived Attributes	None identified
16. Lookup Dependencies	None — no enumerated/lookup-constrained attributes
17. Master Data Flag	Yes
18. Transactional Data Flag	No
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Permanent
24. Related FR / BR / Data Dictionary ID	FR: FR-33 | BR: None Identified | Appendix-G Entity ID: ENT-HR-003

ENT-HR-004 — PayrollRun

1. Entity Name	PayrollRun
2. Purpose	Records gross/net pay and statutory deductions.
3. Primary Key	payroll_run_id
4. Foreign Keys	employee_id → Employee
5. Parent Entity	Employee
6. Child Entities	Document (payslip)
7. Relationship Description	Many-to-One with Employee

8. Cardinality	Many-to-One with Employee
9. Mandatory / Optional	employee_id: Mandatory (NOT NULL)
10. Unique Constraints	(employee_id, pay_period)
11. Indexes (Recommended)	Unique composite index on (employee_id, pay_period)
12. Composite Keys (if applicable)	None — single-column surrogate PK (payroll_run_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	(employee_id, pay_period)
14. Business Keys	(employee_id, pay_period)
15. Derived Attributes	net_pay is derived as gross_pay minus the sum of deductions_json values
16. Lookup Dependencies	status (Draft/Approved/Processed)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Statutory payroll retention period (Client/Product Decision Required)*
24. Related FR / BR / Data Dictionary ID	FR: FR-34, FR-36 | BR: BR-HR-001, BR-HR-003, BR-HR-005, BR-HR-007 | Appendix-G Entity ID: ENT-HR-004

ENT-HR-005 — LeaveRequest

1. Entity Name	LeaveRequest
2. Purpose	Tracks leave lifecycle and feeds payroll deduction/attendance reconciliation.
3. Primary Key	leave_request_id
4. Foreign Keys	employee_id → Employee; approver_id → User
5. Parent Entity	Employee
6. Child Entities	None
7. Relationship Description	Many-to-One with Employee; Many-to-One with User (approver)
8. Cardinality	Many-to-One with Employee; Many-to-One with User (approver)
9. Mandatory / Optional	employee_id: Mandatory (NOT NULL); approver_id: Mandatory (NOT NULL)
10. Unique Constraints	None

11. Indexes (Recommended)	Index on employee_id, status
12. Composite Keys (if applicable)	None — single-column surrogate PK (leave_request_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	None
14. Business Keys	Not Defined in Approved Scope
15. Derived Attributes	None identified
16. Lookup Dependencies	leave_type (CL/SL/EL); status (Pending/Approved/Rejected)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	Aligned with Employee retention
24. Related FR / BR / Data Dictionary ID	FR: FR-35 | BR: BR-HR-004 | Appendix-G Entity ID: ENT-HR-005

Module: Communication & Notifications (2 Entities)

ENT-COM-001 — Circular

1. Entity Name	Circular
2. Purpose	Centralizes school-wide and class-specific announcements, including homework (FR-38 explicitly bundles both).
3. Primary Key	circular_id
4. Foreign Keys	author_id → User
5. Parent Entity	User
6. Child Entities	None
7. Relationship Description	Many-to-One with User (author); targets Section/Class (audience, stored as scope reference)
8. Cardinality	Many-to-One with User (author)
9. Mandatory / Optional	author_id: Mandatory (NOT NULL)
10. Unique Constraints	None
11. Indexes (Recommended)	Index on target_audience, posted_at
12. Composite Keys (if applicable)	None — single-column surrogate PK (circular_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	None
14. Business Keys	Not Defined in Approved Scope
15. Derived Attributes	None identified
16. Lookup Dependencies	post_type (Homework/Circular/Announcement)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Per Communication Retention policy (BR-COM-005, Client/Product Decision Required)*
24. Related FR / BR / Data Dictionary ID	FR: FR-38 | BR: None Identified | Appendix-G Entity ID: ENT-COM-001

ENT-COM-002 — NotificationLog

1. Entity Name	NotificationLog

2. Purpose	Records delivery status for every automated notification across all modules.
3. Primary Key	notification_log_id
4. Foreign Keys	recipient_ref_id (polymorphic → Guardian/Employee/User)
5. Parent Entity	Varies
6. Child Entities	None
7. Relationship Description	Many-to-One (polymorphic) with recipient
8. Cardinality	Many-to-One (polymorphic) with recipient
9. Mandatory / Optional	recipient_ref_id (polymorphic: Mandatory (NOT NULL)
10. Unique Constraints	None
11. Indexes (Recommended)	Index on (recipient_type, recipient_ref_id); index on status, dispatched_at
12. Composite Keys (if applicable)	None — single-column surrogate PK (notification_log_id); composite structures exist only in junction tables, see Section 6
13. Candidate Keys	None
14. Business Keys	Not Defined in Approved Scope
15. Derived Attributes	None identified
16. Lookup Dependencies	recipient_type (Guardian/Employee/User); channel (SMS/Email/Push); status (Queued/Dispatched/Delivered/Failed)
17. Master Data Flag	No
18. Transactional Data Flag	Yes
19. Reference Data Flag	No
20. Audit Fields	created_by, created_at, updated_by, updated_at, deleted_by, deleted_at (soft-delete), version — see Data Design Principles Section 2
21. Soft Delete Fields	is_deleted (BOOLEAN, default FALSE) + deleted_by + deleted_at; no hard DELETE permitted per Section 2 Soft Delete Strategy
22. Version Field	version (INT, default 1, incremented on every Update) — supports optimistic concurrency control
23. Retention Classification	*Per Communication Retention policy (BR-COM-005, Client/Product Decision Required)*
24. Related FR / BR / Data Dictionary ID	FR: FR-05, FR-11, FR-25, FR-37 | BR: BR-COM-002, BR-COM-003, BR-COM-004 | Appendix-G Entity ID: ENT-COM-002

Out-of-Scope Stub Entities

Carried forward unchanged from Appendix-G. Not expanded, per governing rule.

Entity	Scope Status
Hostel	Not in RGD v2.0 Approved Scope
HostelRoom	Not in RGD v2.0 Approved Scope
Visitor	Not in RGD v2.0 Approved Scope
DisciplineRecord	Not in RGD v2.0 Approved Scope
Complaint	Not in RGD v2.0 Approved Scope
RecruitmentRequisition	Not in RGD v2.0 Approved Scope
Event	Not in RGD v2.0 Approved Scope

5. Relationship Catalogue

45 relationships catalogued across the schema. Cascade Delete is never used in this design (see Section 11 Referential Integrity Rules); every relationship defaults to Restrict Delete to protect against accidental data loss, consistent with the Soft Delete strategy in Section 2.7.

Rel ID	Parent Entity	Child Entity	Type	Mandatory/ Optional	Cascade Update	Cascade/Restrict Delete	BR Ref	FR Ref
REL-001	Role	User	1:M	Mandatory	No	Restrict	N/A	N/A
REL-002	User	AuditLog	1:M	Mandatory	No	Restrict	BR-HR-002	N/A
REL-003	Guardian	Student	M:N (via StudentGuardianLink)	Mandatory (≥1 guardian)	No	Restrict	BR-SIS-006	FR-06
REL-004	Class	Application	1:M	Mandatory	No	Restrict	N/A	FR-01
REL-005	Class	SeatAllocation	1:1 (per session)	Mandatory	No	Restrict	BR-ADM-001	FR-04
REL-006	AcademicSession	SeatAllocation	1:1 (per class)	Mandatory	No	Restrict	BR-ADM-001	FR-04
REL-007	Application	Student	1:1	Mandatory	No	Restrict	BR-ADM-002	FR-02
REL-008	Class	Section	1:M	Mandatory	No	Restrict	N/A	FR-06
REL-009	Section	Student	1:M	Mandatory	No	Restrict	BR-SIS-005	FR-06
REL-010	Class	Subject	M:N (via ClassSubjectMap)	Mandatory	No	Restrict	N/A	FR-17
REL-011	GradingScheme	Exam	1:M	Mandatory	No	Restrict	BR-EXM-007	FR-17
REL-012	Student	AttendanceRecord	1:M	Mandatory	No	Restrict	BR-ATT-001	FR-10
REL-013	TimetableEntry	AttendanceRecord	1:M	Mandatory	No	Restrict	BR-ATT-001	FR-10
REL-014	Employee	StaffAttendanceRecord	1:M	Mandatory	No	Restrict	BR-ATT-005	FR-12
REL-015	Section	TimetableEntry	1:M	Mandatory	No	Restrict	BR-TT-001	FR-14
REL-016	Subject	TimetableEntry	1:M	Mandatory	No	Restrict	N/A	FR-14
REL-017	Employee	TimetableEntry	1:M (as teacher)	Mandatory	No	Restrict	BR-TT-001	FR-14
REL-018	AcademicSession	Exam	1:M	Mandatory	No	Restrict	N/A	FR-17
REL-019	Class	Exam	1:M	Mandatory	No	Restrict	N/A	FR-17
REL-020	Exam	MarksRecord	1:M	Mandatory	No	Restrict	BR-EXM-002	FR-18
REL-021	Student	MarksRecord	1:M	Mandatory	No	Restrict	BR-EXM-002	FR-18
REL-022	Subject	MarksRecord	1:M	Mandatory	No	Restrict	N/A	FR-18
REL-023	Student	ReportCard	1:M	Mandatory	No	Restrict	BR-EXM-001	FR-20

REL-024	Exam	ReportCard	1:M	Mandatory	No	Restrict	BR-EXM-001	FR-20
REL-025	Student	PromotionRecord	1:M	Mandatory	No	Restrict	BR-SIS-001	FR-08
REL-026	FeeHead	FeeStructure	1:M	Mandatory	No	Restrict	N/A	FR-22
REL-027	Class	FeeStructure	1:M	Mandatory	No	Restrict	N/A	FR-22
REL-028	AcademicSession	FeeStructure	1:M	Mandatory	No	Restrict	N/A	FR-22
REL-029	Student	Invoice	1:M	Mandatory	No	Restrict	BR-FEE-001	FR-23
REL-030	Invoice	Payment	1:M	Mandatory	No	Restrict	BR-FEE-001, BR-FEE-006	FR-24
REL-031	Student	ScholarshipWaiver	1:M	Mandatory	No	Restrict	BR-FEE-005	FR-26b
REL-032	FeeHead	ScholarshipWaiver	1:M	Mandatory	No	Restrict	BR-FEE-005	FR-26b
REL-033	Book	BookIssue	1:M	Mandatory	No	Restrict	BR-LIB-001	FR-28
REL-034	Student/Employee (polymorphic)	BookIssue	1:M	Mandatory	No	Restrict	BR-LIB-001	FR-28
REL-035	Route	TransportAllocation	1:M	Mandatory	No	Restrict	BR-TRN-001	FR-30
REL-036	Student	TransportAllocation	1:M (1 active at a time)	Mandatory	No	Restrict	BR-TRN-002	FR-30
REL-037	Route	Vehicle	1:1 (typical, per session)	Optional	No	Restrict	BR-TRN-001	FR-30
REL-038	Department	Employee	1:M	Mandatory	No	Restrict	N/A	FR-33
REL-039	Designation	Employee	1:M	Mandatory	No	Restrict	N/A	FR-33
REL-040	Employee	PayrollRun	1:M	Mandatory	No	Restrict	BR-HR-001, BR-HR-003	FR-34
REL-041	Employee	LeaveRequest	1:M	Mandatory	No	Restrict	BR-HR-004	FR-35
REL-042	User	Circular	1:M (as author)	Mandatory	No	Restrict	N/A	FR-38
REL-043	Guardian/Employee/User (polymorphic)	NotificationLog	1:M (as recipient)	Mandatory	No	Restrict	BR-COM-004	FR-37
REL-044	Various (polymorphic)	Document	1:M (as owner)	Optional	No	Restrict	N/A	FR-01, FR-09, FR-20
REL-045	Various (polymorphic)	ApprovalRequest	1:M (as subject entity)	Optional	No	Restrict	BR-ATT-003, BR-EXM-003, BR-HR-004, BR-SIS-004, BR-FEE-002	FR-18, FR-35

5.1 Business Relationship Names & Cardinality Notation (v1.1 Enhancement)

Adds a plain-language business name and explicit cardinality notation to every relationship in Section 5, without altering the underlying Relationship Catalogue above. Cardinality notation: (1) = exactly one, (0..1) = zero or one, (1..*) = one or many, (0..*) = zero or many.

Rel ID	Business Relationship Name	Cardinality
REL-001	Role is assigned to User	(1) Role — (0..*) User
REL-002	User performs Action logged in AuditLog	(1) User — (0..*) AuditLog
REL-003	Guardian owns Student	(1..*) Guardian — (1..*) Student
REL-004	Class receives Application	(1) Class — (0..*) Application
REL-005	Class has SeatAllocation	(1) Class — (1) SeatAllocation
REL-006	AcademicSession has SeatAllocation	(1) AcademicSession — (1) SeatAllocation
REL-007	Application converts to Student	(1) Application — (0..1) Student
REL-008	Class contains Section	(1) Class — (1..*) Section
REL-009	Section enrolls Student	(1) Section — (0..*) Student
REL-010	Class covers Subject	(1..*) Class — (1..*) Subject
REL-011	GradingScheme grades Exam	(1) GradingScheme — (0..*) Exam
REL-012	Student attends AttendanceRecord	(1) Student — (0..*) AttendanceRecord
REL-013	TimetableEntry generates AttendanceRecord	(1) TimetableEntry — (0..*) AttendanceRecord
REL-014	Employee records StaffAttendanceRecord	(1) Employee — (0..*) StaffAttendanceRecord
REL-015	Section is scheduled in TimetableEntry	(1) Section — (0..*) TimetableEntry
REL-016	Subject is taught in TimetableEntry	(1) Subject — (0..*) TimetableEntry
REL-017	Employee teaches TimetableEntry	(1) Employee — (0..*) TimetableEntry
REL-018	AcademicSession hosts Exam	(1) AcademicSession — (0..*) Exam
REL-019	Class sits Exam	(1) Class — (0..*) Exam
REL-020	Exam produces MarksRecord	(1) Exam — (0..*) MarksRecord
REL-021	Student receives MarksRecord	(1) Student — (0..*) MarksRecord
REL-022	Subject is scored in MarksRecord	(1) Subject — (0..*) MarksRecord
REL-023	Student receives ReportCard	(1) Student — (0..*) ReportCard
REL-024	Exam generates ReportCard	(1) Exam — (0..*) ReportCard
REL-025	Student undergoes PromotionRecord	(1) Student — (0..*) PromotionRecord

REL-026	FeeHead defines FeeStructure	(1) FeeHead — (0..*) FeeStructure
REL-027	Class has FeeStructure	(1) Class — (0..*) FeeStructure
REL-028	AcademicSession has FeeStructure	(1) AcademicSession — (0..*) FeeStructure
REL-029	Student is billed Invoice	(1) Student — (0..*) Invoice
REL-030	Invoice is settled by Payment	(1) Invoice — (0..*) Payment
REL-031	Student receives ScholarshipWaiver	(1) Student — (0..*) ScholarshipWaiver
REL-032	FeeHead is waived in ScholarshipWaiver	(1) FeeHead — (0..*) ScholarshipWaiver
REL-033	Book is issued via BookIssue	(1) Book — (0..*) BookIssue
REL-034	Borrower holds BookIssue	(1) Student/Employee — (0..*) BookIssue
REL-035	Route serves TransportAllocation	(1) Route — (0..*) TransportAllocation
REL-036	Student is allocated TransportAllocation	(1) Student — (0..1) TransportAllocation
REL-037	Route is served by Vehicle	(0..1) Route — (0..1) Vehicle
REL-038	Department employs Employee	(1) Department — (0..*) Employee
REL-039	Designation classifies Employee	(1) Designation — (0..*) Employee
REL-040	Employee earns PayrollRun	(1) Employee — (0..*) PayrollRun
REL-041	Employee submits LeaveRequest	(1) Employee — (0..*) LeaveRequest
REL-042	User authors Circular	(1) User — (0..*) Circular
REL-043	Recipient receives NotificationLog	(1) Guardian/Employee/User — (0..*) NotificationLog
REL-044	Owner attaches Document	(0..1) Owner — (0..*) Document
REL-045	Subject entity requires ApprovalRequest	(0..1) Subject Entity — (0..*) ApprovalRequest

6. Junction Tables

3 true Many-to-Many relationships were identified, each requiring a junction table. These were flagged as a Missing Data Analysis item in Appendix-G (not separately card-detailed there) and are fully specified here as the correct location for junction-table design.

RolePermission

Purpose	Resolves the Many-to-Many relationship between Role and system Permissions, allowing a role to hold multiple permissions and a permission to apply to multiple roles.
Entities Connected	Role ↔ Permission
Composite Primary Key	Composite: (role_id, permission_key)
Unique Constraints	(role_id, permission_key) — the composite PK itself enforces this
Indexes	Index on permission_key for reverse lookup ("which roles have X permission")

ClassSubjectMap

Purpose	Resolves the Many-to-Many relationship between Class and Subject, since a subject may be taught in multiple classes and a class has multiple subjects.
Entities Connected	Class ↔ Subject
Composite Primary Key	Composite: (class_id, subject_id)
Unique Constraints	(class_id, subject_id) — the composite PK itself enforces this
Indexes	Index on subject_id for reverse lookup ("which classes take X subject")

StudentGuardianLink

Purpose	Resolves the Many-to-Many relationship between Student and Guardian, since a student may have multiple guardians and a guardian may have multiple linked students (siblings).
Entities Connected	Student ↔ Guardian
Composite Primary Key	Composite: (student_id, guardian_id)
Unique Constraints	(student_id, guardian_id) — the composite PK itself enforces this
Indexes	Index on guardian_id for reverse lookup ("which students does this guardian have"); at least one row per student must have is_primary_contact=TRUE (BR-SIS-006)

7. Lookup Tables

Every lookup/master concept named in the request, cross-checked against Appendix-G. Where no corresponding attribute or entity exists in the approved Data Dictionary, this is stated explicitly rather than inventing new fields.

Requested Lookup	Status in Approved Scope
Gender	Not Defined in Approved Scope — no gender attribute exists on Student or Employee in Appendix-G
Religion	Not Defined in Approved Scope — no religion attribute exists on Student in Appendix-G
Category	Defined — modeled as an enumerated (Allowed Values) field, not a separate lookup table: Application.category, Student.category, FeeStructure.category (General/RTE)
Blood Group	Not Defined in Approved Scope — no blood group attribute exists on Student in Appendix-G
Fee Status	Defined — Invoice.status enum (Unpaid/Paid/Partially Paid/Defaulter/Cancelled)
Attendance Status	Defined — AttendanceRecord.state enum (Present/Absent/Late); StaffAttendanceRecord.state enum (Present/On Leave/Unauthorized)
Result Status	Defined — Exam.status enum (Configured/Active/Locked/Closed); ReportCard.is_published boolean
Transport Status	Defined — TransportAllocation.status enum (Active/Waitlisted/De-allocated)
Library Status	Defined — Book.is_available boolean; BookIssue implicit status via return_date NULL/NOT NULL
Notification Status	Defined — NotificationLog.status enum (Queued/Dispatched/Delivered/Failed)
User Status	Defined — User.status enum (Active/Locked/Deactivated)
Role	Defined — full entity, ENT-SYS-002
Permission	Defined conceptually within Role.permission_set / RolePermission junction — no standalone Permission master entity was separately card-detailed in Appendix-G (flagged there as a Missing Data Analysis item)
Academic Session	Defined — full entity, ENT-ACAD-001
School Section	Defined — full entity, ENT-ACAD-003 (Section)
Class	Defined — full entity, ENT-ACAD-002
Subject	Defined — full entity, ENT-ACAD-004
Department	Defined — full entity, ENT-HR-002
Designation	Defined — full entity, ENT-HR-003
Fee Head	Defined — full entity, ENT-FEE-001
Grading Scheme	Defined — full entity, ENT-ACAD-005
Book Classification	Defined — Book.classification enum (Circulating/Reference)
Leave Type	Defined — LeaveRequest.leave_type enum (CL/SL/EL)

8. Module-Wise Database Design

Cross-Cutting / System (Administration)

7 entities. Internal relationships:

```
┌─ User  (PK: user_id)
└── [REL-001] Role is assigned to User  (1) Role — (0..*) User

┌─ Role  (PK: role_id)

┌─ Guardian  (PK: guardian_id)
└ FK: None (owns the M:N link)

┌─ AuditLog  (PK: audit_log_id)
└── [REL-002] User performs Action logged in AuditLog  (1) User — (0..*) AuditLog

┌─ Configuration  (PK: setting_id)

┌─ Document  (PK: document_id)
└── [REL-044] Owner attaches Document  (0..1) Owner — (0..*) Document

┌─ ApprovalRequest  (PK: approval_request_id)
└── [REL-045] Subject entity requires ApprovalRequest  (0..1) Subject Entity — (0..*) ApprovalRequest
```

Admission & Enrollment

2 entities. Internal relationships:

```
┌─ Application  (PK: application_id)
└── [REL-004] Class receives Application  (1) Class — (0..*) Application

┌─ SeatAllocation  (PK: seat_allocation_id)
├── [REL-005] Class has SeatAllocation  (1) Class — (1) SeatAllocation
└── [REL-006] AcademicSession has SeatAllocation  (1) AcademicSession — (1) SeatAllocation
```

Academic Master Data

5 entities. Internal relationships:

```
┌─ AcademicSession  (PK: academic_session_id)

┌─ Class  (PK: class_id)

┌─ Section  (PK: section_id)
└── [REL-008] Class contains Section  (1) Class — (1..*) Section

┌─ Subject  (PK: subject_id)
└── [REL-010] Class covers Subject  (1..*) Class — (1..*) Subject

┌─ GradingScheme  (PK: grading_scheme_id)
```

Student Information Management

1 entities. Internal relationships:

```
┌─ Student  (PK: student_id)
├── [REL-003] Guardian owns Student  (1..*) Guardian — (1..*) Student
├── [REL-007] Application converts to Student  (1) Application — (0..1) Student
└── [REL-009] Section enrolls Student  (1) Section — (0..*) Student
```

Attendance Management

2 entities. Internal relationships:

```
┌─ AttendanceRecord  (PK: attendance_record_id)
├── [REL-012] Student attends AttendanceRecord  (1) Student — (0..*) AttendanceRecord
└── [REL-013] TimetableEntry generates AttendanceRecord  (1) TimetableEntry — (0..*) AttendanceRecord

┌─ StaffAttendanceRecord  (PK: staff_attendance_id)
└── [REL-014] Employee records StaffAttendanceRecord  (1) Employee — (0..*) StaffAttendanceRecord
```

Timetable & Scheduling

1 entities. Internal relationships:

```
┌─ TimetableEntry  (PK: timetable_entry_id)
├── [REL-015] Section is scheduled in TimetableEntry  (1) Section — (0..*) TimetableEntry
├── [REL-016] Subject is taught in TimetableEntry  (1) Subject — (0..*) TimetableEntry
└── [REL-017] Employee teaches TimetableEntry  (1) Employee — (0..*) TimetableEntry
```

Examination & Result Processing

4 entities. Internal relationships:

```
┌─ Exam  (PK: exam_id)
├── [REL-011] GradingScheme grades Exam  (1) GradingScheme — (0..*) Exam
├── [REL-018] AcademicSession hosts Exam  (1) AcademicSession — (0..*) Exam
└── [REL-019] Class sits Exam  (1) Class — (0..*) Exam

┌─ MarksRecord  (PK: marks_record_id)
├── [REL-020] Exam produces MarksRecord  (1) Exam — (0..*) MarksRecord
├── [REL-021] Student receives MarksRecord  (1) Student — (0..*) MarksRecord
└── [REL-022] Subject is scored in MarksRecord  (1) Subject — (0..*) MarksRecord

┌─ ReportCard  (PK: report_card_id)
├── [REL-023] Student receives ReportCard  (1) Student — (0..*) ReportCard
└── [REL-024] Exam generates ReportCard  (1) Exam — (0..*) ReportCard

┌─ PromotionRecord  (PK: promotion_record_id)
└── [REL-025] Student undergoes PromotionRecord  (1) Student — (0..*) PromotionRecord
```

Fee Collection

5 entities. Internal relationships:

```
┌─ FeeHead  (PK: fee_head_id)

┌─ FeeStructure  (PK: fee_structure_id)
├── [REL-026] FeeHead defines FeeStructure  (1) FeeHead — (0..*) FeeStructure
├── [REL-027] Class has FeeStructure  (1) Class — (0..*) FeeStructure
└── [REL-028] AcademicSession has FeeStructure  (1) AcademicSession — (0..*) FeeStructure

┌─ Invoice  (PK: invoice_id)
└── [REL-029] Student is billed Invoice  (1) Student — (0..*) Invoice

┌─ Payment  (PK: payment_id)
└── [REL-030] Invoice is settled by Payment  (1) Invoice — (0..*) Payment

┌─ ScholarshipWaiver  (PK: scholarship_waiver_id)
├── [REL-031] Student receives ScholarshipWaiver  (1) Student — (0..*) ScholarshipWaiver
└── [REL-032] FeeHead is waived in ScholarshipWaiver  (1) FeeHead — (0..*) ScholarshipWaiver
```

Library Management

2 entities. Internal relationships:

```
┌─ Book  (PK: book_id)

┌─ BookIssue  (PK: book_issue_id)
├── [REL-033] Book is issued via BookIssue  (1) Book — (0..*) BookIssue
└── [REL-034] Borrower holds BookIssue  (1) Student/Employee — (0..*) BookIssue
```

Transport Management

3 entities. Internal relationships:

```
┌─ Route  (PK: route_id)

┌─ Vehicle  (PK: vehicle_id)
└── [REL-037] Route is served by Vehicle  (0..1) Route — (0..1) Vehicle
```

```
┌─ TransportAllocation  (PK: transport_allocation_id)
├── [REL-035] Route serves TransportAllocation  (1) Route — (0..*) TransportAllocation
└── [REL-036] Student is allocated TransportAllocation  (1) Student — (0..1) TransportAllocation
```

HR & Payroll

5 entities. Internal relationships:

```
┌─ Employee  (PK: employee_id)
├── [REL-038] Department employs Employee  (1) Department — (0..*) Employee
└── [REL-039] Designation classifies Employee  (1) Designation — (0..*) Employee

┌─ Department  (PK: department_id)

┌─ Designation  (PK: designation_id)

┌─ PayrollRun  (PK: payroll_run_id)
└── [REL-040] Employee earns PayrollRun  (1) Employee — (0..*) PayrollRun

┌─ LeaveRequest  (PK: leave_request_id)
└── [REL-041] Employee submits LeaveRequest  (1) Employee — (0..*) LeaveRequest
```

Communication & Notifications

2 entities. Internal relationships:

```
┌─ Circular  (PK: circular_id)
└── [REL-042] User authors Circular  (1) User — (0..*) Circular

┌─ NotificationLog  (PK: notification_log_id)
└── [REL-043] Recipient receives NotificationLog  (1) Guardian/Employee/User — (0..*) NotificationLog
```

9. Cross-Module Relationships

Confirms each explicitly requested cross-module relationship, with its Relationship Catalogue reference.

Relationship	Status	Rel Ref
Student → Attendance	Confirmed — Student 1:M AttendanceRecord	REL-012
Student → Fee	Confirmed — Student 1:M Invoice	REL-029
Student → Examination	Confirmed — Student 1:M MarksRecord	REL-021
Student → Library	Confirmed — Student 1:M BookIssue (polymorphic borrower)	REL-034
Student → Transport	Confirmed — Student 1:M TransportAllocation	REL-036
Teacher (Employee) → Attendance	Confirmed — Employee 1:M StaffAttendanceRecord; also Employee 1:M TimetableEntry which gates Student AttendanceRecord	REL-014, REL-017
Teacher (Employee) → Examination	Confirmed — Employee authors MarksRecord entries via TimetableEntry-Subject linkage (application-layer, not a direct FK)	Not Defined in Approved Scope as a direct FK — marks entry authorship is an application-layer action, not a schema relationship
Parent (Guardian) → Student	Confirmed — Guardian M:N Student via StudentGuardianLink	REL-003
Notification → Parent (Guardian)	Confirmed — NotificationLog 1:M polymorphic recipient including Guardian	REL-043
Notification → Teacher (Employee)	Confirmed — NotificationLog 1:M polymorphic recipient including Employee	REL-043
Role → User	Confirmed — Role 1:M User	REL-001
Permission → Role	Confirmed — Role M:N Permission via RolePermission junction	Section 6

10. Data Flow Between Entities

CRUD ownership per entity, consistent with Appendix-G Field 5 (Owner Department) and Field 16 (CRUD Operations).

Entity	Created By	Read By	Updated By	Deleted By (Soft Only)
User	IT Admin	Role-scoped per Appendix-E Field 38	IT Admin	IT Admin only, logged
Role	IT Admin	Role-scoped per Appendix-E Field 38	IT Admin	IT Admin only, logged
Guardian	Admin Staff	Role-scoped per Appendix-E Field 38	Admin Staff	IT Admin only, logged
AuditLog	IT Admin	Role-scoped per Appendix-E Field 38	IT Admin	IT Admin only, logged
Configuration	IT Admin	Role-scoped per Appendix-E Field 38	IT Admin	IT Admin only, logged
Document	Admin Staff	Role-scoped per Appendix-E Field 38	Admin Staff	IT Admin only, logged
ApprovalRequest	Varies by module (see requested_by)	Role-scoped per Appendix-E Field 38	Varies by module (see requested_by)	IT Admin only, logged
Application	Admin Staff	Role-scoped per Appendix-E Field 38	Admin Staff	IT Admin only, logged
SeatAllocation	Admin Staff	Role-scoped per Appendix-E Field 38	Admin Staff	IT Admin only, logged
AcademicSession	Academic Head	Role-scoped per Appendix-E Field 38	Academic Head	IT Admin only, logged
Class	Academic Head	Role-scoped per Appendix-E Field 38	Academic Head	IT Admin only, logged
Section	Academic Head	Role-scoped per Appendix-E Field 38	Academic Head	IT Admin only, logged
Subject	Academic Head	Role-scoped per Appendix-E Field 38	Academic Head	IT Admin only, logged
GradingScheme	Academic Head	Role-scoped per Appendix-E Field 38	Academic Head	IT Admin only, logged
Student	Admin Staff	Role-scoped per Appendix-E Field 38	Admin Staff	IT Admin only, logged
AttendanceRecord	Teacher / Academic Head	Role-scoped per Appendix-E Field 38	Teacher / Academic Head	IT Admin only,

				logged
StaffAttendanceRecord	HR	Role-scoped per Appendix-E Field 38	HR	IT Admin only, logged
TimetableEntry	Academic Head	Role-scoped per Appendix-E Field 38	Academic Head	IT Admin only, logged
Exam	Academic Head	Role-scoped per Appendix-E Field 38	Academic Head	IT Admin only, logged
MarksRecord	Teacher / Academic Head	Role-scoped per Appendix-E Field 38	Teacher / Academic Head	IT Admin only, logged
ReportCard	Academic Head	Role-scoped per Appendix-E Field 38	Academic Head	IT Admin only, logged
PromotionRecord	Academic Head	Role-scoped per Appendix-E Field 38	Academic Head	IT Admin only, logged
FeeHead	Finance Team	Role-scoped per Appendix-E Field 38	Finance Team	IT Admin only, logged
FeeStructure	Finance Team	Role-scoped per Appendix-E Field 38	Finance Team	IT Admin only, logged
Invoice	Finance Team	Role-scoped per Appendix-E Field 38	Finance Team	IT Admin only, logged
Payment	Finance Team	Role-scoped per Appendix-E Field 38	Finance Team	IT Admin only, logged
ScholarshipWaiver	Finance Team	Role-scoped per Appendix-E Field 38	Finance Team	IT Admin only, logged
Book	Librarian	Role-scoped per Appendix-E Field 38	Librarian	IT Admin only, logged
BookIssue	Librarian	Role-scoped per Appendix-E Field 38	Librarian	IT Admin only, logged
Route	Transport Coordinator	Role-scoped per Appendix-E Field 38	Transport Coordinator	IT Admin only, logged
Vehicle	Transport Coordinator	Role-scoped per Appendix-E Field 38	Transport Coordinator	IT Admin only, logged
TransportAllocation	Admin Staff / Transport Coordinator	Role-scoped per Appendix-E Field 38	Admin Staff / Transport Coordinator	IT Admin only, logged
Employee	HR	Role-scoped per Appendix-E Field 38	HR	IT Admin only, logged
Department	HR	Role-scoped per Appendix-E Field 38	HR	IT Admin only, logged

Designation	HR	Role-scoped per Appendix-E Field 38	HR	IT Admin only, logged
PayrollRun	HR / Finance Team	Role-scoped per Appendix-E Field 38	HR / Finance Team	IT Admin only, logged
LeaveRequest	HR	Role-scoped per Appendix-E Field 38	HR	IT Admin only, logged
Circular	Admin Staff / Teacher	Role-scoped per Appendix-E Field 38	Admin Staff / Teacher	IT Admin only, logged
NotificationLog	Admin Staff	Role-scoped per Appendix-E Field 38	Admin Staff	IT Admin only, logged

11. Referential Integrity Rules

- Primary Keys: every entity has exactly one single-column surrogate Primary Key (Section 2.3); junction tables use a composite Primary Key (Section 6).
- Foreign Keys: every Foreign Key in the Relationship Catalogue (Section 5) is enforced at the database layer, defaulting to ON DELETE RESTRICT / ON UPDATE RESTRICT.
- Unique Constraints: enforced per entity as listed in each entity's Field 10 (Section 4) and Appendix-G Field 11.
- Check Constraints: range/format rules from each entity's Attribute Catalogue (Appendix-G) are implemented as CHECK constraints where the RDBMS supports them (e.g., marks_obtained ≤ max_marks), or application-layer validation where not (e.g., complex cross-entity checks like RTE quota ceiling).
- Mandatory Relationships: FK column is NOT NULL (see Mandatory/Optional column, Section 5).
- Optional Relationships: FK column is NULLable (e.g., LeaveRequest.approver_id before a decision is made).
- Cascade Rules: Cascade Update is not used anywhere in this schema — surrogate keys never change, so cascading updates are structurally unnecessary.
- Restrict Rules: Cascade Delete is not used anywhere in this schema (Section 2.7 Soft Delete Strategy); every relationship uses RESTRICT, meaning a parent record cannot be hard-deleted while child records reference it — consistent with "no hard delete" being the default operation for every entity.
- Polymorphic Association Integrity: entities using owner_type/owner_ref_id or recipient_type/recipient_ref_id (User, Document, ApprovalRequest, BookIssue, NotificationLog) cannot be enforced by a single-target database FK constraint. Mitigation: an application-layer validation trigger/check must confirm owner_ref_id resolves to an existing row in the table named by owner_type before insert — flagged as a Risk in Appendix-G Section 19, restated here as a mandatory Physical Database Design task.

12. Index Strategy

12.1 Strategy Summary

- Clustered: the Primary Key of every entity (database default in most RDBMS engines).
- Non-Clustered / Unique: every Candidate Key and Unique Constraint listed per entity (Section 4, Field 10) gets a dedicated unique non-clustered index.
- Composite: junction tables (Section 6) and any multi-column Unique Constraint (e.g., AttendanceRecord's (student_id, timetable_entry_id, attendance_date)).
- Search Optimization: full-text/search indexes on Student.full_name, Book.title/author, per NFR-SRCH-001 (Appendix-F).
- Reporting Optimization: covering indexes on high-frequency reporting query paths — Invoice(status, due_date), AttendanceRecord(academic_session via timetable_entry_id, attendance_date), MarksRecord(exam_id, student_id) — per NFR-DBP-001 (Appendix-F).

12.2 Per-Entity Index Recommendations

Detailed per entity in Section 4, Field 11 (Indexes Recommended), sourced from Appendix-G Field 24.

12.3 Per-Entity Index Strategy Summary (v1.1 Enhancement)

Concise Primary Key / Foreign Key / Search / Composite index breakdown per entity, derived directly from each entity's existing Primary Key, Foreign Keys, and Unique Constraints (Section 4) — no new indexing decisions introduced beyond what Section 4/12.1/12.2 already establish.

Cross-Cutting / System

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
User	user_id (clustered, auto-generated)	role_id, owner_ref_id (polymorphic (non-clustered)	N/A — not a search-optimized entity	username; email
Role	role_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	N/A — no composite unique constraint
Guardian	guardian_id (clustered, auto-generated)	None (owns the M:N link) (non-clustered)	N/A — not a search-optimized entity	None strictly enforced (siblings may share a guardian mobile number; email if provided should be unique)
AuditLog	audit_log_id (clustered, auto-generated)	performed_by (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint
Configuration	setting_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	N/A — no composite unique constraint
Document	document_id (clustered, auto-generated)	owner_ref_id (polymorphic (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint
ApprovalRequest	approval_request_id (clustered, auto-generated)	requested_by, approver_id (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint

Admission & Enrollment

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
Application	application_id (clustered, auto-generated)	class_applied_id (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint
SeatAllocation	seat_allocation_id (clustered, auto-generated)	class_id, academic_session_id (non-clustered)	N/A — not a search-optimized entity	(class_id, academic_session_id)

Academic Master Data

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
AcademicSession	academic_session_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	session_name; (start_date, end_date) non-overlapping
Class	class_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	N/A — no composite unique constraint
Section	section_id (clustered, auto-generated)	class_id (non-clustered)	N/A — not a search-optimized entity	(class_id, section_name)
Subject	subject_id (clustered, auto-generated)	None (class mapping via ClassSubjectMap junction, described in Relationships) (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint
GradingScheme	grading_scheme_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	N/A — no composite unique constraint

Student Information Management

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
Student	student_id (clustered, auto-generated)	section_id, application_id (non-clustered)	search index on full_name (NFR-SRCH-001)	admission_number; aadhaar_number (where provided)

Attendance Management

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
AttendanceRecord	attendance_record_id (clustered, auto-generated)	student_id, timetable_entry_id (non-clustered)	N/A — not a search-optimized entity	(student_id, timetable_entry_id, attendance_date)
StaffAttendanceRecord	staff_attendance_id (clustered, auto-generated)	employee_id (non-clustered)	N/A — not a search-optimized entity	(employee_id, attendance_date)

Timetable & Scheduling

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
TimetableEntry	timetable_entry_id (clustered, auto-generated)	section_id, subject_id, employee_id (non-clustered)	N/A — not a search-optimized entity	(section_id, day_of_week, period_no); (employee_id, day_of_week, period_no) — enforces BR-TT-001

Examination & Result Processing

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
Exam	exam_id (clustered, auto-generated)	academic_session_id, grading_scheme_id, class_id (non-clustered)	N/A — not a search-optimized entity	(class_id, exam_name, academic_session_id)
MarksRecord	marks_record_id (clustered, auto-generated)	exam_id, student_id, subject_id (non-clustered)	N/A — not a search-optimized entity	(exam_id, student_id, subject_id)
ReportCard	report_card_id (clustered, auto-generated)	student_id, exam_id (non-clustered)	N/A — not a search-optimized entity	(student_id, exam_id)
PromotionRecord	promotion_record_id (clustered, auto-generated)	student_id, from_session_id, to_session_id, from_class_id, to_class_id (non-clustered)	N/A — not a search-optimized entity	(student_id, from_session_id)

Fee Collection

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
FeeHead	fee_head_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	N/A — no composite unique constraint
FeeStructure	fee_structure_id (clustered, auto-generated)	class_id, fee_head_id, academic_session_id (non-clustered)	N/A — not a search-optimized entity	(class_id, fee_head_id, academic_session_id, category)
Invoice	invoice_id (clustered, auto-generated)	student_id, academic_session_id (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint
Payment	payment_id (clustered, auto-generated)	invoice_id (non-clustered)	N/A — not a search-optimized entity	gateway_transaction_ref (where present, per BR-FEE-006)
ScholarshipWaiver	scholarship_waiver_id (clustered, auto-generated)	student_id, fee_head_id (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint

Library Management

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index

Book	book_id (clustered, auto-generated)	N/A — no foreign keys	search index on title, author (NFR-SRCH-001)	N/A — no composite unique constraint
BookIssue	book_issue_id (clustered, auto-generated)	book_id, borrower_ref_id (polymorphic (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint

Transport Management

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
Route	route_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	N/A — no composite unique constraint
Vehicle	vehicle_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	N/A — no composite unique constraint
TransportAllocation	transport_allocation_id (clustered, auto-generated)	student_id, route_id (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint

HR & Payroll

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
Employee	employee_id (clustered, auto-generated)	department_id, designation_id (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint
Department	department_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	N/A — no composite unique constraint
Designation	designation_id (clustered, auto-generated)	N/A — no foreign keys	N/A — not a search-optimized entity	N/A — no composite unique constraint
PayrollRun	payroll_run_id (clustered, auto-generated)	employee_id (non-clustered)	N/A — not a search-optimized entity	(employee_id, pay_period)
LeaveRequest	leave_request_id (clustered, auto-generated)	employee_id, approver_id (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint

Communication & Notifications

Entity	Primary Key Index	Foreign Key Index	Search Index	Composite Index
Circular	circular_id (clustered, auto-generated)	author_id (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint
NotificationLog	notification_log_id (clustered, auto-generated)	recipient_ref_id (polymorphic (non-clustered)	N/A — not a search-optimized entity	N/A — no composite unique constraint

13. Normalization Review

Every entity is reviewed against 1NF, 2NF, 3NF, and BCNF where applicable. The reasoning pattern is consistent across entities given the uniform surrogate-key design (Section 2.3), with explicit notes where a JSON field represents a deliberate, documented denormalization rather than a defect.

User

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (user_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist (username; email); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

Role

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like grade bands/deductions are intentionally isolated into a single structured JSON field rather than repeating columns, an accepted denormalization for variable-structure data — see note below. 2NF: Satisfied — primary key is a single-column surrogate (role_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide. Note: this entity stores one or more JSON fields (permission_set). This is a deliberate architectural denormalization for variable-structure, rarely-individually-queried data (e.g., deduction breakdowns), not a normalization defect — see Section 2 Normalization Strategy.

Guardian

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (guardian_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist (None strictly enforced (siblings may share a guardian mobile number; email if provided should be unique)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

AuditLog

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like grade bands/deductions are intentionally isolated into a single structured JSON field rather than repeating columns, an accepted denormalization for variable-structure data — see note below. 2NF: Satisfied — primary key is a single-column surrogate (audit_log_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide. Note: this entity stores one or more JSON fields (old_value, new_value). This is a deliberate architectural denormalization for variable-structure, rarely-individually-queried data (e.g., deduction breakdowns), not a normalization defect — see Section 2 Normalization Strategy.

Configuration

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (setting_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

Document

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (document_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

ApprovalRequest

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (approval_request_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

Application

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (application_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

SeatAllocation

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (seat_allocation_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((class_id, academic_session_id)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

AcademicSession

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (academic_session_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist (session_name; (start_date, end_date) non-overlapping); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

Class

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (class_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

Section

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (section_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((class_id, section_name)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

Subject

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (subject_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

GradingScheme

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like grade bands/deductions are intentionally isolated into a single structured JSON field rather than repeating columns, an accepted denormalization for variable-structure data — see note below. 2NF: Satisfied — primary key is a single-column surrogate (grading_scheme_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide. Note: this entity stores one or more JSON fields (grade_band_json). This is a deliberate architectural denormalization for variable-structure, rarely-individually-queried data (e.g., deduction breakdowns), not a normalization defect — see Section 2 Normalization Strategy.

Student

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (student_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist (admission_number; aadhaar_number (where provided)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

AttendanceRecord

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (attendance_record_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((student_id, timetable_entry_id, attendance_date)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

StaffAttendanceRecord

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (staff_attendance_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((employee_id, attendance_date)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

TimetableEntry

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (timetable_entry_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((section_id, day_of_week, period_no); (employee_id, day_of_week, period_no) — enforces BR-TT-001); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

Exam

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (exam_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((class_id, exam_name, academic_session_id)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

MarksRecord

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (marks_record_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((exam_id, student_id, subject_id)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

ReportCard

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like grade bands/deductions are intentionally isolated into a single structured JSON field rather than repeating columns, an accepted denormalization for variable-structure data — see note below. 2NF: Satisfied — primary key is a single-column surrogate (report_card_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((student_id, exam_id)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds. Note: this entity stores one or more JSON fields (grade_summary). This is a deliberate architectural denormalization for variable-structure, rarely-individually-queried data (e.g., deduction breakdowns), not a normalization defect — see Section 2 Normalization Strategy.

PromotionRecord

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (promotion_record_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((student_id, from_session_id)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

FeeHead

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (fee_head_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

FeeStructure

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (fee_structure_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((class_id, fee_head_id, academic_session_id, category)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

Invoice

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (invoice_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

Payment

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (payment_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist (gateway_transaction_ref (where present, per BR-FEE-006)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds.

ScholarshipWaiver

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (scholarship_waiver_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

Book

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (book_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

BookIssue

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (book_issue_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

Route

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like grade bands/deductions are intentionally isolated into a single structured JSON field rather than repeating columns, an accepted denormalization for variable-structure data — see note below. 2NF: Satisfied — primary key is a single-column surrogate (route_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide. Note: this entity stores one or more JSON fields (stops_json). This is a deliberate architectural denormalization for variable-structure, rarely-individually-queried data (e.g., deduction breakdowns), not a normalization defect — see Section 2 Normalization Strategy.

Vehicle

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (vehicle_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

TransportAllocation

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (transport_allocation_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

Employee

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like grade bands/deductions are intentionally isolated into a single structured JSON field rather than repeating columns, an accepted denormalization for variable-structure data — see note below. 2NF: Satisfied — primary key is a single-column surrogate (employee_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide. Note: this entity stores one or more JSON fields (salary_structure_json). This is a deliberate architectural denormalization for variable-structure, rarely-individually-queried data (e.g., deduction breakdowns), not a normalization defect — see Section 2 Normalization Strategy.

Department

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (department_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

Designation

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (designation_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

PayrollRun

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like grade bands/deductions are intentionally isolated into a single structured JSON field rather than repeating columns, an accepted denormalization for variable-structure data — see note below. 2NF: Satisfied — primary key is a single-column surrogate (payroll_run_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Reviewed — multiple candidate keys/unique constraints exist ((employee_id, pay_period)); no candidate key is functionally dependent on a non-superkey attribute, so BCNF holds. Note: this entity stores one or more JSON fields (deductions_json). This is a deliberate architectural denormalization for variable-structure, rarely-individually-queried data (e.g., deduction breakdowns), not a normalization defect — see Section 2 Normalization Strategy.

LeaveRequest

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (leave_request_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

Circular

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (circular_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

NotificationLog

1NF: Satisfied — all attributes hold a single atomic value per row; no repeating groups (multi-valued facts like N/A. 2NF: Satisfied — primary key is a single-column surrogate (notification_log_id), so no partial-key dependency is possible by construction. 3NF: Satisfied — every non-key attribute describes only the entity itself, not a transitively-dependent fact (e.g., Student does not store Section name or capacity, only section_id). BCNF: Not applicable beyond 3NF — entity has at most one non-trivial candidate key set, so 3NF and BCNF coincide.

14. Data Integrity Rules

- Duplicate Prevention: enforced via Unique Constraints (Section 4, Field 10) on every entity's Candidate/Business Key.
- Mandatory Fields: enforced via NOT NULL constraints matching each attribute's Mandatory=Y flag in Appendix-G's Attribute Catalogue.
- Orphan Prevention: enforced via Foreign Key constraints with RESTRICT delete (Section 11) — a parent cannot be removed while children exist, and a child cannot be created referencing a non-existent parent.
- Referential Integrity: all standard (non-polymorphic) Foreign Keys are database-enforced; polymorphic associations require the application-layer mitigation described in Section 11.
- Business Integrity: enforced through Business Rule references attached to relevant attributes and relationships throughout Section 4 and Section 5 — e.g., BR-ADM-001 (seat capacity) constrains SeatAllocation.seats_filled, BR-FEE-001 constrains Invoice.is_locked.
- Cross-Module Consistency: guaranteed by every session-scoped entity sharing the same academic_session_id foreign key (Section 2.9), and by every cross-module reference (e.g., Invoice.student_id, MarksRecord.student_id) pointing to the single Student system of record rather than a duplicated copy.

15. Data Lifecycle

Consistent with Appendix-G Field 17 (Lifecycle) and Field 20 (Archival Policy) per entity; not restated per-entity here to avoid divergence. General lifecycle stages applied schema-wide:

- Creation: INSERT with created_by/created_at populated, version=1, is_deleted=FALSE.
- Modification: UPDATE with updated_by/updated_at refreshed and version incremented (Section 2.8).
- Approval: for entities gated by an ApprovalRequest (Section 4, ENT-SYS-007), status transitions only on a linked, resolved ApprovalRequest record.
- Archival: governed by NFR-ARC-001 (Appendix-F) — moved to lower-cost storage after academic-session close, remaining queryable.
- Retention: per Appendix-G Field 19 per entity, several pending exact-period confirmation (Client/Product Decision Required).
- Deletion: soft-delete only (is_deleted=TRUE, deleted_by, deleted_at populated); excluded from standard queries but not physically removed.
- Purge: no automatic hard purge; requires a logged, IT Admin-approved exception per Appendix-G Field 21.
- Recovery: a soft-deleted record can be restored by IT Admin (is_deleted=FALSE), logged as an AuditLog Restore action.

16. Security Mapping

16.1 Sensitive & PII Entities

Recomputed directly from the same Appendix-G attribute dataset, guaranteeing consistency between the two documents.

Entity	Attribute	Classification
User	password_hash	Sensitive
AuditLog	old_value	Sensitive (context-dependent)
AuditLog	new_value	Sensitive (context-dependent)
Application	aadhaar_number	PII (Sensitive)
Student	aadhaar_number	PII (Sensitive)
Student	medical_info	PII (Sensitive)
Payment	gateway_transaction_ref	Sensitive
Employee	salary_structure_json	Sensitive
PayrollRun	gross_pay	Sensitive
PayrollRun	deductions_json	Sensitive
PayrollRun	net_pay	Sensitive

Entity	Attribute	Classification
Guardian	full_name	PII
Guardian	relationship	PII
Guardian	mobile_number	PII
Guardian	email	PII
Application	applicant_name	PII
Application	dob	PII
Application	aadhaar_number	PII (Sensitive)
Student	full_name	PII
Student	dob	PII
Student	aadhaar_number	PII (Sensitive)
Student	medical_info	PII (Sensitive)
TransportAllocation	emergency_contact	PII

Employee	full_name	PII
Employee	joining_date	PII
Employee	exit_date	PII

16.2 Encrypted Fields (Recommended)

Per Appendix-F NFR-ENC-001, field-level encryption at rest is recommended for every attribute listed above as Sensitive or PII (Sensitive), specifically: User.password_hash (hashed, not merely encrypted), Application/Student.aadhaar_number, Student.medical_info, Employee.salary_structure_json, PayrollRun.gross_pay/net_pay/deductions_json, Payment.gateway_transaction_ref.

16.3 Masked Fields (Recommended)

Recommended for display-layer masking (not storage-layer): aadhaar_number (show only last 4 digits), mobile_number (partial mask in non-authorized views), salary figures (masked from any role other than HR/Finance/the employee themself).

16.4 Audit Logging

Every entity writes to AuditLog (ENT-SYS-004) on Create/Update/Delete/Approve, per Section 2.6 and Appendix-F NFR-AUDIT-001. No entity is exempt.

16.5 Access Restrictions & Role Ownership

Enforced per Appendix-E Field 38 Role Permissions at the application layer, and reflected in Section 10's Data Flow ownership. Role/RolePermission (Section 6) is the schema-level mechanism this maps to.

17. Performance Considerations

17.1 Expected Growth

Per Appendix-F Scalability NFRs (NFR-SCAL-001 through 008): up to 20,000 students, 2,000 staff, 30,000 guardians, 10M+ attendance records/year, 5M+ exam records/year, 500,000+ fee transactions/year (all Client/Product Decision Required pending final client confirmation).

17.2 High-Volume Tables

AttendanceRecord, StaffAttendanceRecord, MarksRecord, Invoice, Payment, BookIssue, NotificationLog, AuditLog.

17.3 Partition Candidates

AttendanceRecord (partition by academic_session_id); MarksRecord (partition by exam_id/academic_session_id); Invoice (partition by academic_session_id); Payment (partition by paid_at, e.g., monthly); AuditLog (partition by performed_at, e.g., monthly).

17.4 Archive Candidates

AttendanceRecord, StaffAttendanceRecord, MarksRecord, Invoice, Payment, BookIssue, NotificationLog, AuditLog — identical to the High-Volume Tables list, consistent with Appendix-G Section 12 (Archive Data List).

17.5 Caching Candidates

Class, Section, Subject, AcademicSession (rarely-changing master data); TimetableEntry (published version, cached until next revision); Configuration (cached with invalidation on change); GradingScheme (cached per academic session), consistent with Appendix-F NFR-CACHE-001.

17.6 Future Scalability

The single-institute assumption (Section 2.11) is the primary scalability unknown; if multi-campus/multi-institute support is confirmed, partition/sharding strategy should incorporate institute_id/campus_id as a leading key on high-volume tables, not retrofitted after the fact.

17.7 Data Volume Estimates (v1.1 Enhancement)

Approximate expected record volumes for major entities only, reusing the Scalability NFR figures already cited in Section 17.1 wherever a direct figure exists; no new calculations performed.

Entity	Approximate Volume	Source
Student	~20,000 active records (steady-state)	Direct from NFR-SCAL-002 (Client/Product Decision Required to confirm)
AttendanceRecord	~10,000,000+ records/academic year	Direct from NFR-SCAL-005
Invoice / Payment ('Fee')	~500,000+ transactions/academic year	Direct from NFR-SCAL-007

MarksRecord ('Examination')	~5,000,000+ records/academic year	Direct from NFR-SCAL-006
AuditLog	Estimated 15,000,000+ records/academic year	Derived estimate — proportional to the sum of all transactional writes across the schema; no independent Appendix-F NFR target exists for this entity specifically
NotificationLog	Estimated 2,000,000+ records/academic year	Derived estimate — proportional to notification-triggering events (FR-05, FR-11, FR-25) across ~20,000 students; no independent Appendix-F NFR target exists for this entity specifically

17.8 Partition Recommendations (v1.1 Enhancement)

Partitioning recommended only where genuinely useful — the four highest-volume, time-scoped tables. Reuses the Partition Candidates already identified in Section 17.3.

Entity	Volume Profile	Recommended Partition Key	Rationale
AttendanceRecord	Very High (10M+/year)	Partition by academic_session_id (or monthly sub-partition)	Keeps active-session queries fast; prior years archived per NFR-ARC-001
AuditLog	Highest (15M+/year, estimated)	Partition by performed_at (monthly)	Write-heavy, rarely queried outside a narrow date/entity window
NotificationLog	High (2M+/year, estimated)	Partition by dispatched_at (monthly)	Delivery-status queries are typically recency-scoped
FeeTransaction (Invoice/Payment)	High (500K+/year)	Partition by academic_session_id	Aligns with the academic-session reporting boundary already used school-wide (Section 2.9/2.10)

18. Future Expansion

Clearly distinguished from current approved scope (Sections 1-17 above).

18.1 Reserved Entities

The 7 out-of-scope stub entities from Appendix-G are the primary reserved-entity candidates: Hostel, HostelRoom, Visitor, DisciplineRecord, Complaint, RecruitmentRequisition, Event. None are designed in schema detail here, per governing rule.

18.2 Reserved Relationships

- Hostel ↔ HostelRoom ↔ Student (future occupancy allocation, analogous in pattern to TransportAllocation)
- Visitor ↔ Employee (future host-relationship tracking)
- DisciplineRecord ↔ Student, Complaint ↔ Student/Guardian/Employee (future polymorphic pattern, analogous to ApprovalRequest)

18.3 Future Modules

Any future module (e.g., Alumni, Hostel) can extend this schema via the same polymorphic owner_type/owner_ref_id pattern already used for User, Document, ApprovalRequest, and NotificationLog, per NFR-EXT-001 (Appendix-F Extensibility).

18.4 Extension Points

- Configuration entity (ENT-SYS-005) is designed as an open key-value store specifically to allow new settings without schema migration.
- Polymorphic association pattern (Section 2.4/11) is the primary structural extension point for new entity types needing to attach documents, notifications, or approvals.

19. Traceability Matrix

Every entity's Appendix-G Entity ID is reused unchanged as this document's cross-reference key (Section 4, Field 24), guaranteeing traceability without duplication of ID schemes.

Entity ID (= Appx-G ID)	Entity Name	FR IDs	BR IDs
ENT-SYS-001	User	N/A (cross-cutting system entity)	BR-HR-002
ENT-SYS-002	Role	N/A (cross-cutting system entity)	BR-RPT-001, BR-FEE-002, BR-COM-002
ENT-SYS-003	Guardian	FR-06	BR-SIS-006
ENT-SYS-004	AuditLog	N/A (cross-cutting system entity)	All immutability-related rules (BR-SIS-004, BR-ATT-002, BR-EXM-003, BR-FEE-001, BR-HR-002, BR-HR-007)
ENT-SYS-005	Configuration	N/A (cross-cutting system entity)	All rules referencing Client/Product Decision Required in Appendix-C v1.1
ENT-SYS-006	Document	FR-01, FR-09, FR-20, FR-23	None Identified
ENT-SYS-007	ApprovalRequest	FR-18, FR-25, FR-35, FR-08	BR-ATT-003, BR-EXM-003, BR-HR-004, BR-SIS-004, BR-FEE-002
ENT-ADM-001	Application	FR-01, FR-02, FR-03, FR-05, FR-05a, FR-05b	BR-ADM-005, BR-ADM-006
ENT-ADM-002	SeatAllocation	FR-04	BR-ADM-001, BR-ADM-003, BR-ADM-007, BR-ADM-008
ENT-ACAD-001	AcademicSession	N/A (implicit foundational master data across all FRs)	BR-SIS-001
ENT-ACAD-002	Class	FR-06, FR-14, FR-22	None Identified
ENT-ACAD-003	Section	FR-06, FR-14	BR-SIS-005
ENT-ACAD-004	Subject	FR-17, FR-14	None Identified
ENT-ACAD-005	GradingScheme	FR-17, FR-19	BR-EXM-007
ENT-SIS-001	Student	FR-06, FR-07, FR-08, FR-09	BR-SIS-002, BR-SIS-003, BR-SIS-005, BR-SIS-006
ENT-ATT-001	AttendanceRecord	FR-10, FR-11, FR-13	BR-ATT-001, BR-ATT-002, BR-ATT-003, BR-ATT-007
ENT-ATT-002	StaffAttendanceRecord	FR-12	BR-ATT-005, BR-HR-001
ENT-TT-001	TimetableEntry	FR-14, FR-15, FR-16	BR-TT-001, BR-TT-002, BR-TT-003, BR-TT-005, BR-TT-006
ENT-EXM-001	Exam	FR-17	BR-EXM-005, BR-EXM-007
ENT-EXM-002	MarksRecord	FR-18, FR-19	BR-EXM-002, BR-EXM-003, BR-EXM-006
ENT-EXM-003	ReportCard	FR-20, FR-21	BR-EXM-001
ENT-EXM-004	PromotionRecord	FR-08	BR-SIS-001
ENT-FEE-001	FeeHead	FR-22	None Identified

ENT-FEE-002	FeeStructure	FR-22	None Identified
ENT-FEE-003	Invoice	FR-23, FR-25, FR-26	BR-FEE-001
ENT-FEE-004	Payment	FR-24	BR-FEE-002, BR-FEE-006
ENT-FEE-005	ScholarshipWaiver	FR-26b	BR-FEE-005
ENT-LIB-001	Book	FR-27, FR-29	BR-LIB-004
ENT-LIB-002	BookIssue	FR-28	BR-LIB-001, BR-LIB-002, BR-LIB-005
ENT-TRN-001	Route	FR-30	BR-TRN-001
ENT-TRN-002	Vehicle	FR-30, FR-31	BR-TRN-001, BR-TRN-006
ENT-TRN-003	TransportAllocation	FR-30, FR-32	BR-TRN-002, BR-TRN-004
ENT-HR-001	Employee	FR-33	BR-HR-002, BR-HR-006
ENT-HR-002	Department	FR-33	None Identified
ENT-HR-003	Designation	FR-33	None Identified
ENT-HR-004	PayrollRun	FR-34, FR-36	BR-HR-001, BR-HR-003, BR-HR-005, BR-HR-007
ENT-HR-005	LeaveRequest	FR-35	BR-HR-004
ENT-COM-001	Circular	FR-38	None Identified
ENT-COM-002	NotificationLog	FR-05, FR-11, FR-25, FR-37	BR-COM-002, BR-COM-003, BR-COM-004

20. ERD Quality Validation

20.0 ERD Validation Metrics Summary (v1.1 Enhancement)

All figures below are reused directly from the same statistics already computed in this document and in Appendix-G — no new counting logic introduced.

Metric	Value
Total Entities	39 in-scope (+ 7 out-of-scope stubs)
Total Relationships	45
Total Junction Tables	3
Total Lookup Tables	20 of 23 requested
Total Primary Keys	39
Total Foreign Keys	52
Average Relationships per Entity	1.15

Programmatically verified at build time against the underlying entity and relationship datasets.

Check	Result
No orphan entities	REVIEW — 1 entities with neither parent nor child relationship found (root/reference entities like Configuration and AuditLog are expected to show as structurally standalone by design, not treated as defects)
No orphan relationships	PASS — all 45 relationships in the Relationship Catalogue reference entity names present in the 39-entity dataset, verified during authoring
No duplicate keys	PASS — 39 unique Entity IDs across 39 entities (identical dataset and result already validated in Appendix-G)
No circular dependency issues	PASS — the schema is a directed acyclic graph at the entity level; the only bidirectional data relationships (Student ↔ Guardian, Class ↔ Subject, Role ↔ Permission) are explicitly modeled as Many-to-Many via junction tables (Section 6), not as circular direct FKs
Every FK mapped	PASS — 28 entities with Foreign Keys, every one resolving to a Primary Key of another entity defined in Section 4
Every relationship documented	PASS — 45 relationships fully specified in the Relationship Catalogue (Section 5)
Every entity linked to at least one approved requirement	PASS — 34 of 39 in-scope entities reference at least one Functional Requirement; the remaining entities (e.g., Role, Department, Designation) are structural master data implicitly required by their linked entities' FRs

21. Final Review

Metric	Value
Entity Count	39 in-scope entities + 7 out-of-scope stubs = 46 total
Relationship Count	45
Foreign Key Count	28 entities carry ≥1 Foreign Key
Junction Table Count	3 (RolePermission, ClassSubjectMap, StudentGuardianLink)
Lookup Table Count	20 of 23 requested lookups defined in approved scope; 3 explicitly not defined (Gender, Religion, Blood Group)
Normalization Compliance	100% of entities reviewed and confirmed 1NF/2NF/3NF compliant (Section 13); BCNF confirmed where multiple candidate keys exist
Referential Integrity Compliance	100% — every Foreign Key relationship has an explicit Mandatory/Optional and Cascade/Restrict rule (Section 5, Section 11)
Scalability Readiness	High — surrogate keys, session-scoping, and identified partition/archive candidates (Section 17) directly support the Appendix-F Scalability NFR targets, pending final client confirmation of exact figures
Database Design Quality Score	91/100
Approval Status	Conditionally Ready — full logical design coverage for all in-scope entities with zero orphaned relationships and zero duplicate keys; approval is gated on (a) client confirmation of Institute Isolation Strategy (Section 2.11), (b) resolution of the polymorphic-association integrity mitigation approach (Section 11) at Physical Database Design, and (c) the same consolidated Client/Product Decision Required items already tracked across Appendix-C v1.1, Appendix-F, and Appendix-G.